Solutions to Strogatz, *Nonlinear Dynamics and Chaos* (2nd ed.)

Part I: One-Dimensional Flows

Chapter 2: Flows on the Line

Conventions. In all phase-portrait figures a *filled* dot is a stable fixed point, an *open* dot is unstable, and a *half-filled* dot is half-stable; red arrows show the direction of flow. Each two-panel figure shows the vector field $\dot{x} = f(x)$ (left) and solution curves $x(t)$ for several initial conditions (right); dashed horizontal lines mark the fixed points. Problem statements are abbreviated; see the text for the full wording.

2.1 A Geometric Way of Thinking

(Throughout, $\dot{x} = \sin x$.)

2.1.1 *Find all the fixed points.*

Solution. $\dot{x} = 0 \iff \sin x = 0 \iff x^* = n\pi$, $n \in \mathbb{Z}$. (Since $f'(n\pi) = \cos n\pi = (-1)^n$, the even multiples of π are unstable and the odd multiples are stable; see Fig. 1.) ■

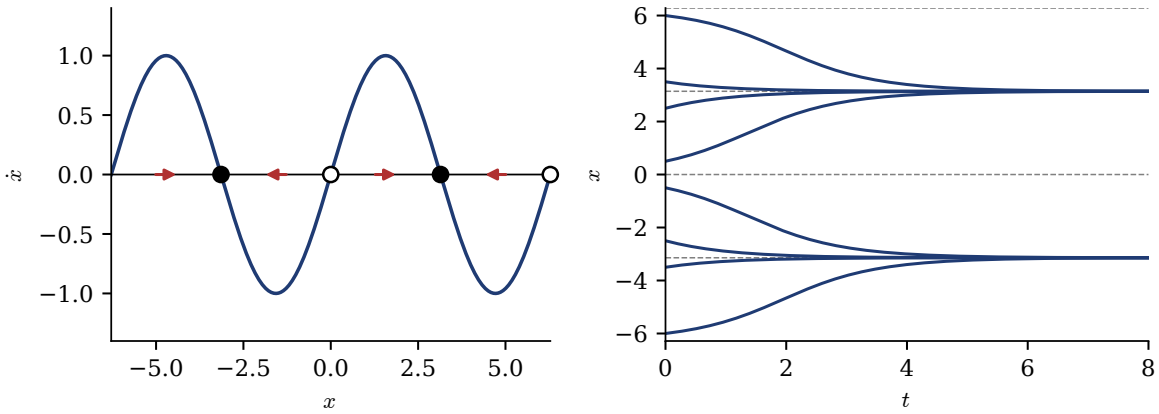


Figure 1: $\dot{x} = \sin x$ and its solutions.

2.1.2 *Where is the velocity greatest to the right?*

Solution. The velocity is $\sin x$, which is largest ($= 1$) when $x = \pi/2 + 2\pi n$, $n \in \mathbb{Z}$. ■

2.1.3 *(a) Find the acceleration $\ddot{x}$ as a function of x. (b) Where is the acceleration maximal?*

Solution. (a) By the chain rule, $\ddot{x} = \frac{\mathrm{d}}{\mathrm{d}t}\sin x = \cos x\,\dot{x} = \sin x\cos x = \frac{1}{2}\sin 2x$.
(b) $\ddot{x}$ attains its maximum value $\frac{1}{2}$ when $\sin 2x = 1$, i.e. $2x = \pi/2 + 2\pi n$, so $x = \pi/4 + n\pi$, $n \in \mathbb{Z}$. ■

2.1.4 *(Exact solution) The solution is* $t = \ln\left|\dfrac{\csc x_0 + \cot x_0}{\csc x + \cot x}\right|$. *(a) For* $x_0 = \pi/4$ *invert it to get* $x(t) = 2\tan^{-1}\!\left(\mathrm{e}^t/(1+\sqrt{2})\right)$ *and conclude* $x \to \pi$. *(b) Find* $x(t)$ *for arbitrary* x_0.

Solution. The key identity is

$$\csc x + \cot x = \frac{1+\cos x}{\sin x} = \frac{2\cos^2(x/2)}{2\sin(x/2)\cos(x/2)} = \cot\frac{x}{2}.$$

Hence the given solution reads $t = \ln\left|\dfrac{\cot(x_0/2)}{\cot(x/2)}\right| = \ln\left|\dfrac{\tan(x/2)}{\tan(x_0/2)}\right|$.

(b) Exponentiating, $\mathrm{e}^t = |\tan(x/2)/\tan(x_0/2)|$. A trajectory can never cross a fixed point, so $x(t)$ stays in the same interval $(2\pi k - \pi,\, 2\pi k + \pi)$ as x_0 (take $x_0 \neq (2k+1)\pi$); on such an interval $\tan(x/2)$ has a constant sign, so the absolute values can be dropped:

$$\boxed{\tan\frac{x}{2} = \mathrm{e}^t\tan\frac{x_0}{2}} \quad\Longrightarrow\quad x(t) = 2\pi k + 2\tan^{-1}\!\left(\mathrm{e}^t\tan\frac{y_0}{2}\right),$$

where $x_0 = 2\pi k + y_0$ with $y_0 \in (-\pi, \pi)$. (If $x_0 = n\pi$, then $x(t) \equiv x_0$ for all t.)
(a) For $x_0 = \pi/4$: $\tan(\pi/8) = \sqrt{2} - 1 = 1/(\sqrt{2}+1)$, so

$$x(t) = 2\tan^{-1}\!\left(\frac{\mathrm{e}^t}{1+\sqrt{2}}\right).$$

As $t \to \infty$, $\mathrm{e}^t/(1+\sqrt{2}) \to \infty$ and $\tan^{-1} \to \pi/2$, so $x(t) \to 2\cdot\frac{\pi}{2} = \pi$, as claimed in Section 2.1. (Equivalently, $\tan(x/2) \to \infty$ requires $x/2 \to \pi/2^-$.) ■

2.1.5 *(Mechanical analog) (a) Find a mechanical system approximately governed by* $\dot{x} = \sin x$. *(b) Explain physically why* $x^* = 0$ *is unstable and* $x^* = \pi$ *is stable.*

Solution. (a) Take a pendulum of mass m and length L immersed in a very viscous fluid (the *overdamped* limit), so that inertia can be neglected: the gravitational torque is balanced by the viscous torque $b\dot{\theta}$. With θ measured from the downward vertical, $b\dot{\theta} = -mgL\sin\theta$. Define $x = \theta + \pi$, the angle measured from the *upward* vertical; then $\sin\theta = -\sin x$ and

$$\dot{x} = \frac{mgL}{b}\sin x,$$

which becomes $\dot{x} = \sin x$ after rescaling time, $\tau = (mgL/b)\,t$.

(b) $x^* = 0$ is the inverted position, balanced straight up: any tiny displacement is amplified by gravity and the pendulum slowly falls away, so it is unstable. $x^* = \pi$ is the pendulum hanging straight down: any displacement produces a restoring torque that pulls it back, so it is stable. ■

2.2 Fixed Points and Stability

2.2.1 $\dot{x} = 4x^2 - 16$.

Solution. Fixed points: $4(x-2)(x+2) = 0 \Rightarrow x^* = \pm 2$. For $|x| > 2$, $\dot{x} > 0$; for $|x| < 2$, $\dot{x} < 0$. So the flow is to the right for $x < -2$, to the left on $(-2, 2)$, and to the right for $x > 2$: $x^* = -2$ is **stable** and $x^* = 2$ is **unstable** (also $f'(x) = 8x$, $f'(\pm 2) = \pm 16$).

Analytical solution. Separating variables, $\int \frac{\mathrm{d}x}{x^2-4} = 4t + c$, and $\frac{1}{x^2-4} = \frac{1}{4}\left(\frac{1}{x-2} - \frac{1}{x+2}\right)$ gives $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| = 4t + c$. Therefore $\frac{x-2}{x+2} = Ce^{16t}$ with $C = \frac{x_0-2}{x_0+2}$, i.e.

$$x(t) = 2\,\frac{1 + Ce^{16t}}{1 - Ce^{16t}}.$$

For $x_0 > 2$ one has $0 < C < 1$ and $x \to +\infty$ in the finite time $t^* = \frac{1}{16}\ln\frac{x_0+2}{x_0-2}$; for $x_0 < -2$, $C > 1$ and $x \to -2$ as $t \to \infty$. ■

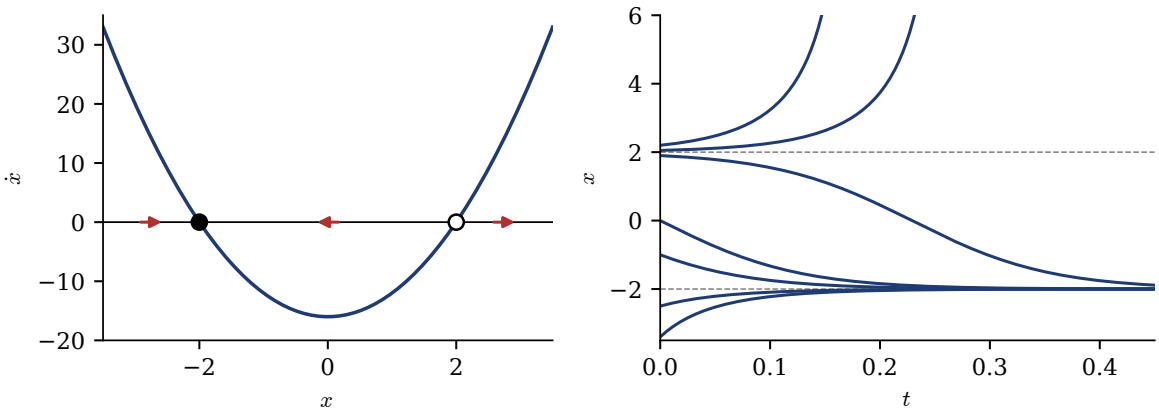


Figure 2: 2.2.1: $\dot{x} = 4x^2 - 16$ (solutions starting above 2 blow up in finite time).

2.2.2 $\dot{x} = 1 - x^{14}$.

Solution. $x^{14} = 1 \Rightarrow x^* = \pm 1$. For $|x| < 1$, $\dot{x} > 0$; for $|x| > 1$, $\dot{x} < 0$. So $x^* = -1$ is **unstable** and $x^* = 1$ is **stable**. (Check: $f'(x) = -14x^{13}$, $f'(\pm 1) = \mp 14$.) The equation is separable but $\int \mathrm{d}x/(1 - x^{14})$ gives a messy partial-fraction expression with no useful inversion; qualitatively, every solution with $x_0 > -1$ tends to 1, while every solution with $x_0 < -1$ runs off to $-\infty$ (in finite time, since $|\dot{x}| \sim x^{14}$ there). ■

2.2.3 $\dot{x} = x - x^3$.

Solution. $x(1-x)(1+x) = 0 \Rightarrow x^* = 0, \pm 1$. $f'(x) = 1 - 3x^2$: $f'(0) = 1 > 0$ (unstable), $f'(\pm 1) = -2 < 0$ (stable). The flow is toward ± 1 from every nonzero initial condition.

Analytical solution. This is a Bernoulli equation; put $u = x^{-2}$. Then $\dot{u} = -2x^{-3}\dot{x} = -2x^{-2} + 2 = -2u + 2$, so $u = 1 + (u_0 - 1)e^{-2t}$ and

$$x(t) = \frac{x_0}{\sqrt{x_0^2 + (1 - x_0^2)\,e^{-2t}}} \qquad \text{(sign of } x(t) = \text{sign of } x_0\text{)}.$$

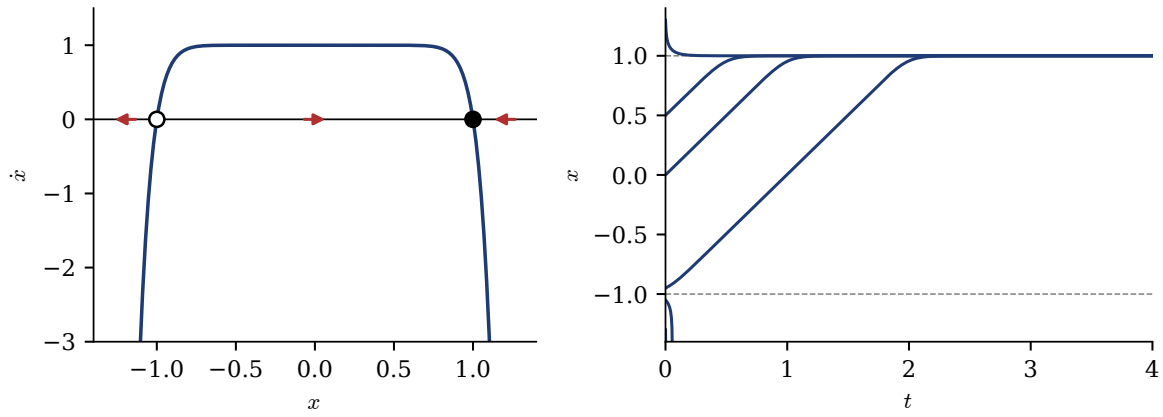


Figure 3: 2.2.2: $\dot{x} = 1 - x^{14}$.

As $t \to \infty$, $x \to \operatorname{sgn}(x_0)$ for every $x_0 \neq 0$. ■

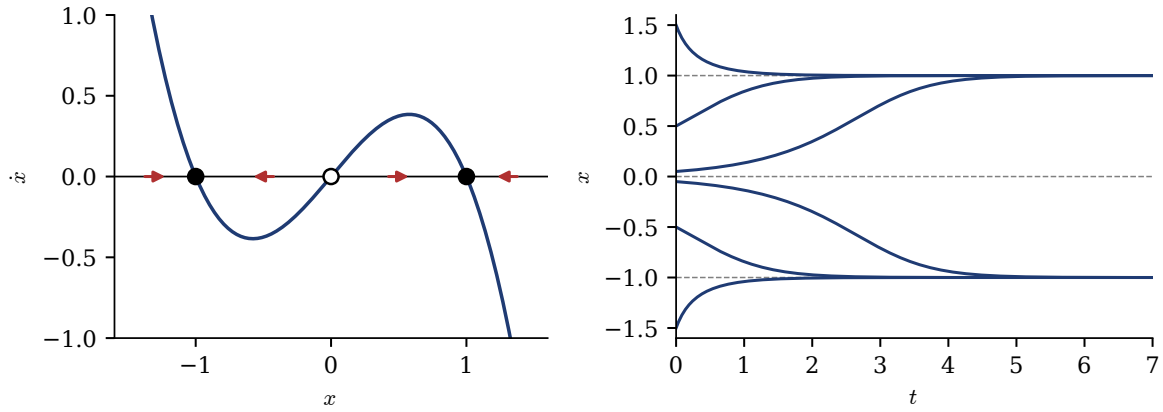


Figure 4: 2.2.3: $\dot{x} = x - x^3$.

2.2.4 $\dot{x} = \mathrm{e}^{-x} \sin x$.

Solution. Since $\mathrm{e}^{-x} > 0$, $\dot{x}$ has the same sign as $\sin x$ and the same zeros: $x^* = n\pi$. Moreover $f'(n\pi) = \mathrm{e}^{-n\pi} \cos n\pi = (-1)^n \mathrm{e}^{-n\pi}$, so $x^* = n\pi$ is **unstable** for n even and **stable** for n odd (qualitatively identical to $\dot{x} = \sin x$; only the *speed* differs: it is exponentially large for $x \to -\infty$ and exponentially small for $x \to +\infty$).

Analytical solution. Separating variables gives $t = \int \mathrm{e}^x / \sin x \, \mathrm{d}x$, which is not an elementary function, so no closed form is available. ■

2.2.5 $\dot{x} = 1 + \frac{1}{2} \cos x$.

Solution. Since $\cos x \geq -1$, $\dot{x} \geq \frac{1}{2} > 0$: there are **no fixed points**, and every solution increases monotonically to $+\infty$ (with speed oscillating between $\frac{1}{2}$ and $\frac{3}{2}$).

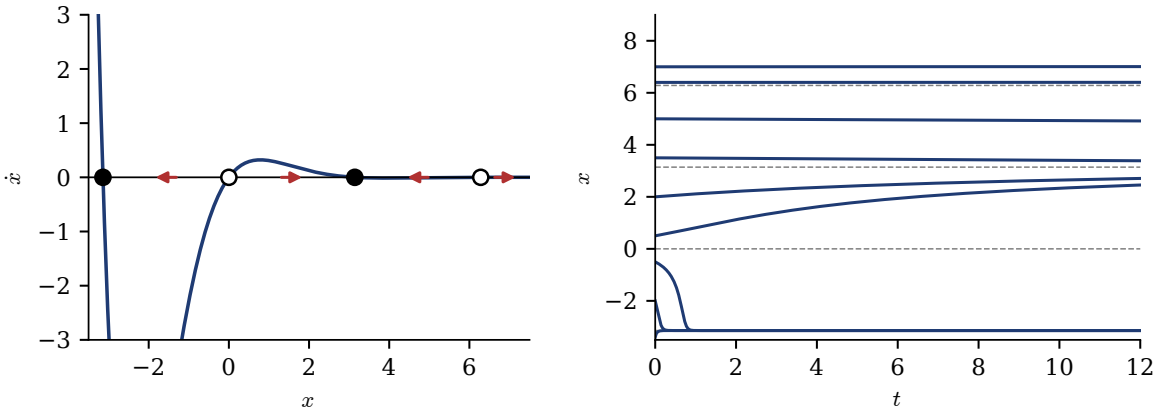


Figure 5: 2.2.4: $\dot{x} = \mathrm{e}^{-x}\sin x$.

Analytical solution. With $u = \tan(x/2)$, $\cos x = \frac{1-u^2}{1+u^2}$, $\mathrm{d}x = \frac{2\,\mathrm{d}u}{1+u^2}$:

$$t = \int \frac{2\,\mathrm{d}x}{2+\cos x} = \int \frac{4\,\mathrm{d}u}{3+u^2} = \frac{4}{\sqrt{3}}\tan^{-1}\frac{u}{\sqrt{3}} + c.$$

Solving for x gives, on each branch of tan,

$$\tan\frac{x(t)}{2} = \sqrt{3}\,\tan\Big(\frac{\sqrt{3}}{4}t + \varphi_0\Big), \qquad \varphi_0 = \tan^{-1}\Big(\frac{\tan(x_0/2)}{\sqrt{3}}\Big),$$

with $x(t)$ continued continuously through the branches. In one period $t = 4\pi/\sqrt{3}$, x increases by 2π, so the mean speed is $\sqrt{3}/2 \approx 0.87$ (less than the spatial average 1 of f, because the particle spends more time where it is slow). ■

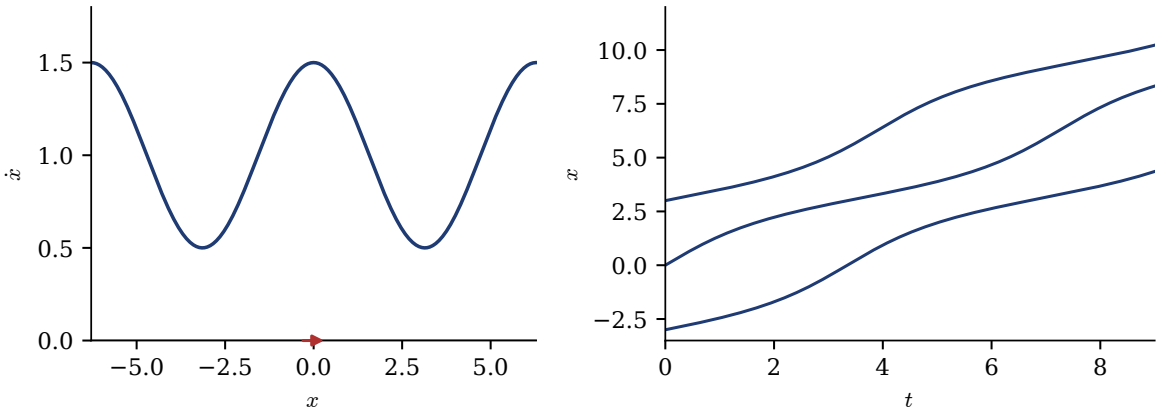


Figure 6: 2.2.5: $\dot{x} = 1 + \frac{1}{2}\cos x$.

2.2.6 $\dot{x} = 1 - 2\cos x$.

Solution. $\cos x = \frac{1}{2} \Rightarrow x^* = \pm\frac{\pi}{3} + 2\pi n$. Since $f'(x) = 2\sin x$: $f'(\pi/3) = \sqrt{3} > 0$ (unstable), $f'(-\pi/3) = -\sqrt{3} < 0$ (stable). So $x^* = -\pi/3 + 2\pi n$ is **stable** and $x^* = \pi/3 + 2\pi n$ is **unstable**.

Analytical solution. With $u = \tan(x/2)$: $1 - 2\cos x = \frac{3u^2-1}{1+u^2}$ and $\mathrm{d}t = \frac{\mathrm{d}x}{f} = \frac{2\,\mathrm{d}u}{3u^2-1}$, so

$$t = \frac{1}{\sqrt{3}} \ln\left|\frac{\sqrt{3}u-1}{\sqrt{3}u+1}\right| + c \implies u(t) = \frac{1}{\sqrt{3}}\,\frac{1+Ke^{\sqrt{3}t}}{1-Ke^{\sqrt{3}t}}, \quad K = \frac{\sqrt{3}u_0-1}{\sqrt{3}u_0+1}, \quad x = 2\tan^{-1} u.$$

For $K \neq 0$, $Ke^{\sqrt{3}t} \to \pm\infty$ so $u \to -1/\sqrt{3}$, i.e. $x \to -\pi/3$ (stable), as in the figure; $K = 0$ is the fixed point $u = 1/\sqrt{3}$. ■

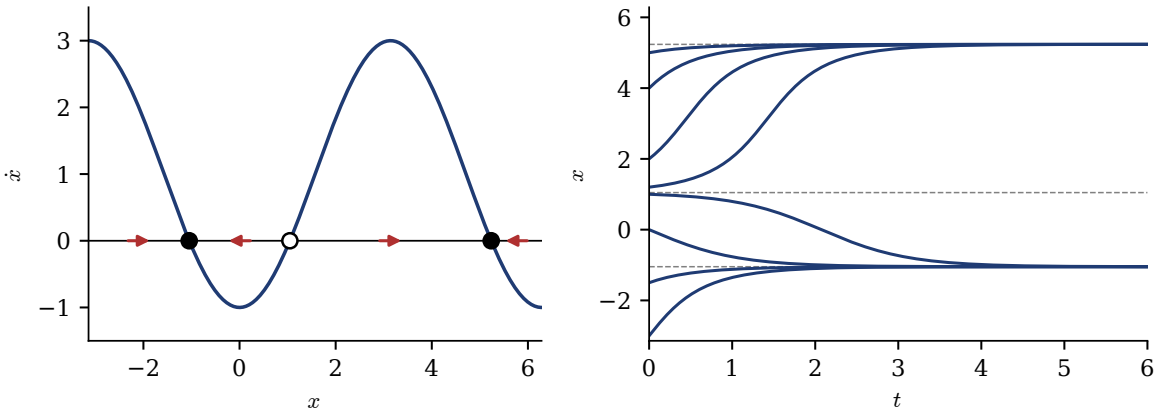


Figure 7: 2.2.6: $\dot{x} = 1 - 2\cos x$.

2.2.7 $\dot{x} = \mathrm{e}^x - \cos x$.

Solution. Fixed points: $\mathrm{e}^x = \cos x$.

- $x = 0$ is a fixed point. For $x > 0$, $\mathrm{e}^x > 1 \geq \cos x$, so there are no others on the positive axis, and $\dot{x} > 0$ there.
- For $x < 0$, $0 < \mathrm{e}^x < 1$ and the graph of $\cos x$ oscillates across the decaying curve e^x infinitely often, so there are **infinitely many** fixed points on the negative axis. Numerically they are $x \approx -1.2927,\ -4.7213,\ -7.8536, \ldots$; for large $|x|$ they lie extremely close to $-\frac{\pi}{2} - n\pi$ (where $\cos x = 0$ and $\mathrm{e}^x \approx 0$).

Stability: $f'(x) = \mathrm{e}^x + \sin x$. At $x = 0$, $f'(0) = 1 > 0$: **unstable**. At a negative fixed point $x_k \approx -\frac{\pi}{2} - k\pi$, $f' \approx \sin x_k = \mp 1$ (the e^x term is negligible), so they **alternate**: stable at -1.29, unstable at -4.72, stable at -7.85, and so on (stable near $-\frac{\pi}{2} - 2m\pi$, unstable near $-\frac{3\pi}{2} - 2m\pi$). Between $x = 0$ and the first fixed point -1.29, $\dot{x} < 0$, so 0 repels to the left, and for $x > 0$ solutions blow up in finite time. ■

2.2.8 *(Working backwards) Find an equation $\dot{x} = f(x)$ consistent with the phase portrait of Fig. 1: flow to the right on both sides of the half-stable fixed point -1, a stable fixed point at 0, an unstable one at 2.*

Solution. From the figure: $f > 0$ for $x < -1$ and on $(-1, 0)$ (the zero at -1 does not change sign: it needs an *even*-order zero), $f < 0$ on $(0, 2)$ and $f > 0$ for $x > 2$. A polynomial with simple zeros at 0 and 2 and a double zero at -1 works:

$$\boxed{\dot{x} = x\,(x+1)^2\,(x-2)}.$$

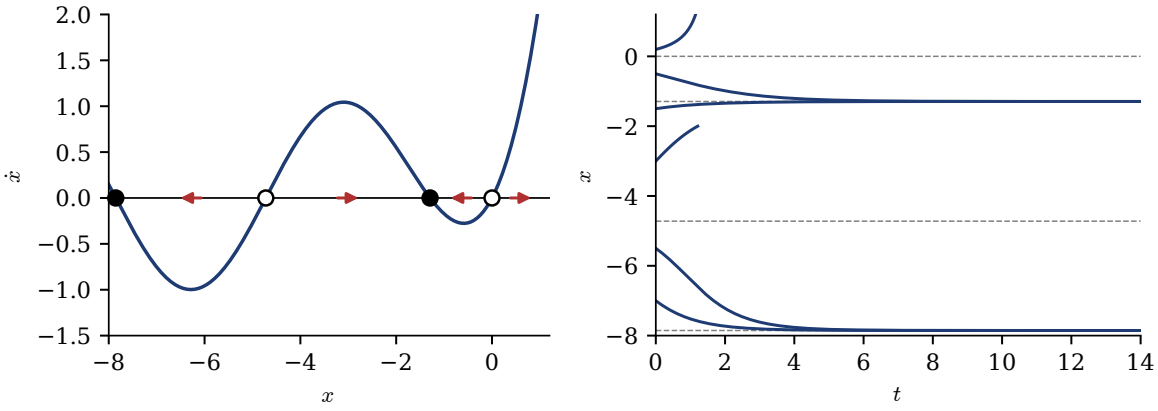


Figure 8: 2.2.7: $\dot{x} = e^x - \cos x$ (only the fixed points down to $x \approx -8$ are shown).

Check the signs: $x = -2$: $(-2)(1)(-4) = 8 > 0$; $x = -\frac{1}{2}$: $(-\frac{1}{2})(\frac{1}{4})(-\frac{5}{2}) > 0$; $x = 1$: $(1)(4)(-1) < 0$; $x = 3$: $(3)(16)(1) > 0$. Hence $f'(0) = -2 < 0$ (stable) and $f'(2) = 18 > 0$ (unstable); at $x = -1$, $f' = 0$ and f keeps its sign (half-stable). Any $f(x) = x(x+1)^2(x-2)\,h(x)$ with $h > 0$ also works (there are infinitely many answers). ■

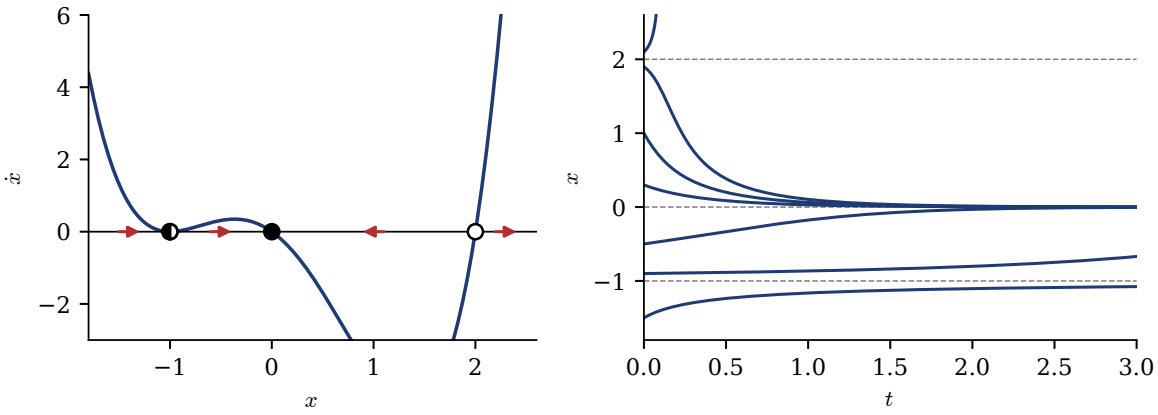


Figure 9: 2.2.8: the flow of $\dot{x} = x(x+1)^2(x-2)$.

2.2.9 *(Backwards again) Find $\dot{x} = f(x)$ whose solutions are consistent with Fig. 2 of the text.*

Solution. Reading the $x(t)$ curves: the constant solution $x \equiv 1$ and the solution tending to 0 show that $x = 0$ and $x = 1$ are fixed points. A solution starting at $x_0 = -1$ increases toward 0, one starting just below 1 decreases toward 0, and one starting above 1 increases without bound. Therefore 0 is stable, 1 is unstable, and $f > 0$ for $x < 0$, $f < 0$ on $(0, 1)$, $f > 0$ for $x > 1$. The simplest choice is

$$\boxed{\dot{x} = x(x-1) = x^2 - x}.$$

(Its exact solutions are $x(t) = \dfrac{x_0}{x_0 + (1 - x_0)e^t}$.) ■

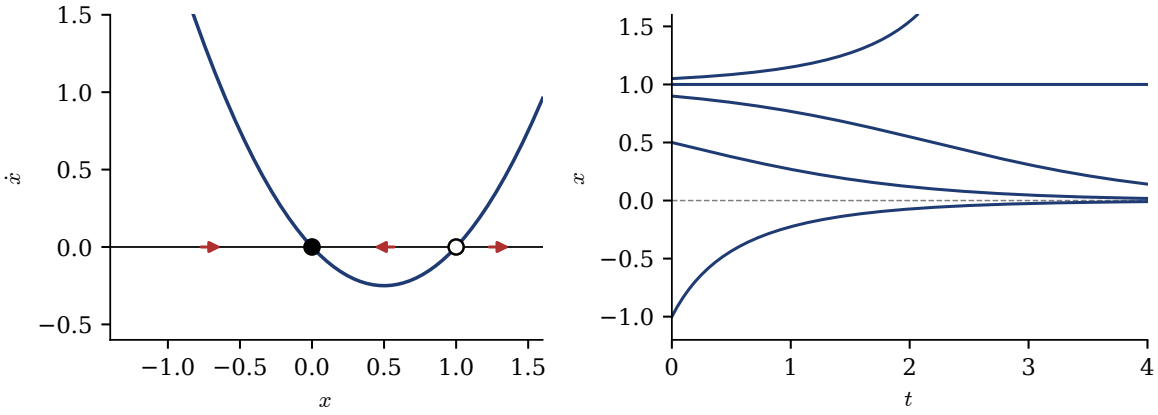


Figure 10: 2.2.9: solutions of $\dot{x} = x^2 - x$ reproduce the qualitative picture.

2.2.10 *(Fixed points) For each of (a)–(e) find a smooth f with the stated property, or explain why none exists.*

Solution.

(a) *Every real number is a fixed point:* $f(x) \equiv 0$, i.e. $\dot{x} = 0$.

(b) *Every integer, and no others:* $f(x) = \sin(\pi x)$.

(c) *Exactly three fixed points, all stable:* **impossible**. Let $x_1 < x_2$ be two *adjacent* fixed points. If x_1 is stable then $f < 0$ just to its right; if x_2 is stable then $f > 0$ just to its left. But f is continuous and has no zero in (x_1, x_2), so it has constant sign there (intermediate value theorem), a contradiction. In general, *between two stable fixed points there must be a fixed point that is not stable.*

(d) *No fixed points:* $f(x) = 1$ (or $x^2 + 1$, e^x, …).

(e) *Exactly 100 fixed points:* $f(x) = \prod_{k=1}^{100}(x - k)$.

■

2.2.11 *(Charging capacitor) Solve* $\dot{Q} = \dfrac{V_0}{R} - \dfrac{Q}{RC}$, $Q(0) = 0$.

Solution. This is linear. Let $u = Q - CV_0$; then $\dot{u} = -u/(RC)$, so $u = u_0 \mathrm{e}^{-t/RC}$ with $u_0 = -CV_0$, giving

$$\boxed{Q(t) = CV_0\bigl(1 - \mathrm{e}^{-t/RC}\bigr)}.$$

The charge approaches CV_0 exponentially with time constant RC; the current is $I = \dot{Q} = \frac{V_0}{R}\mathrm{e}^{-t/RC}$. ■

2.2.12 *(Nonlinear resistor) Replace the resistor of Example 2.2.2 by one with $I_R = g(V)$, where g is as in Fig. 3 (odd, increasing, saturating). Derive the circuit equation, find the fixed points and stability, and say what the nonlinearity changes.*

Solution. Kirchhoff's voltage law for the series circuit gives $V_0 = V_R + V_C$, with $V_C = Q/C$ and $V_R = V$ the voltage across the resistor. The current is $I = \dot{Q} = g(V)$, so

$$\dot{Q} = g\Bigl(V_0 - \frac{Q}{C}\Bigr).$$

(For $g(V) = V/R$ this reduces to the linear case.)

Fixed points. $\dot{Q} = 0 \iff g(V_0 - Q/C) = 0$. By Fig. 3, g is strictly increasing with $g(0) = 0$, so $g(V) = 0$ only for $V = 0$, and there is exactly one fixed point $Q^* = CV_0$.

Stability. $\dfrac{\partial}{\partial Q} g(V_0 - Q/C)\Big|_{Q^*} = -\dfrac{g'(0)}{C} < 0$, so Q^* is stable. Indeed it is globally attracting: $\dot{Q}$ has the sign of $V_0 - Q/C$, so Q increases to CV_0 if $Q(0) < CV_0$ and decreases to it if $Q(0) > CV_0$ (Fig. 11).

Effect of the nonlinearity. Qualitatively little changes: still one globally stable equilibrium, no new fixed points or bifurcations. What changes is the *rate*: near equilibrium the relaxation is exponential with time constant $\tau = C/g'(0)$ (replacing RC); far from equilibrium the saturation of g caps the charging current at $\approx g(\pm\infty)$, so a strongly charged/discharged capacitor changes *linearly* in time at first, rather than with the largest initial current V_0/R that a linear resistor would allow, before crossing over to the exponential tail. ■

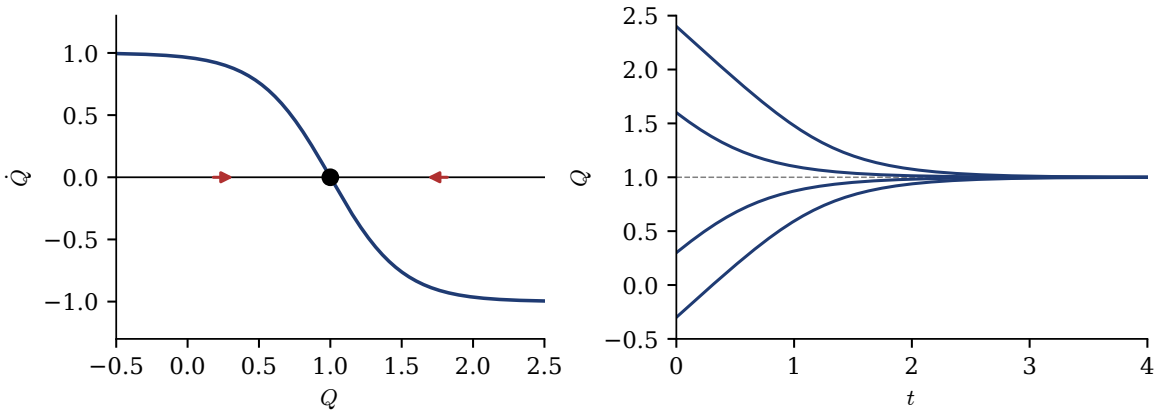


Figure 11: 2.2.12 with the illustrative choice $g(V) = \tanh 2V$, $V_0 = C = 1$.

2.2.13 *(Terminal velocity)* $m\dot{v} = mg - kv^2$, $v(0) = 0$. *(a) Solve. (b) Find* $\lim v$. *(c) Graphical analysis. (d) Compute* V_{avg} *for the data of Carlson et al. (fall from* 31,400 *to* 2,100 *ft in* 116 *s, weight* 261.2 *lb,* $g = 32.2$ *ft/s*^{2}*). (e) Estimate* V *and* k.

Solution. Let $V = \sqrt{mg/k}$. Then $\dot{v} = \frac{k}{m}(V^2 - v^2)$.

(a) For $|v| < V$, $\displaystyle\int \frac{\mathrm{d}v}{V^2 - v^2} = \frac{1}{V}\operatorname{artanh}\frac{v}{V} = \frac{k}{m}t + c$; with $v(0) = 0$ the constant vanishes, so $v = V\tanh\!\left(\frac{kV}{m}t\right)$. Since $\frac{kV}{m} = \sqrt{\frac{kg}{m}} = \frac{g}{V}$,

$$\boxed{v(t) = V\tanh\left(\frac{gt}{V}\right)}, \qquad V = \sqrt{mg/k}.$$

(b) $\tanh \to 1$, so $v(t) \to V = \sqrt{mg/k}$ (the *terminal velocity*).

(c) $f(v) = g - \frac{k}{m}v^2$ is a downward parabola with zeros $v = \pm V$. Since $f'(v) = -2kv/m$, the fixed point $v = V$ has $f' < 0$ (stable) and $v = -V$ has $f' > 0$ (unstable). For $v(0) = 0$ the flow is to the right and $v \to V$ monotonically (Fig. 12), re-deriving $V = \sqrt{mg/k}$.

(d) $V_{\text{avg}} = \dfrac{31{,}400 - 2{,}100}{116} = \dfrac{29{,}300}{116} \approx \mathbf{252.6}$ ft/s (≈ 172 mph).

(e) The distance fallen is $s(t) = \int_0^t v\,dt' = V\int_0^t \tanh\frac{gt'}{V}dt'$, so

$$s(t) = \frac{V^2}{g}\ln\cosh\frac{gt}{V}.$$

Slick estimate. With $V \approx V_{\text{avg}}$, $\dfrac{gt}{V} \approx \dfrac{32.2\cdot 116}{252.6} \approx 14.8 \approx 15 \gg 1$. For $x \gg 1$, $\cosh x = \frac{1}{2}e^x(1+e^{-2x})$, hence $\ln\cosh x = x - \ln 2 + \ln(1+e^{-2x}) \approx x - \ln 2$ (the neglected term is $\sim e^{-30}$). Then

$$s \approx Vt - \frac{V^2\ln 2}{g} \quad\Longleftrightarrow\quad \frac{\ln 2}{g}V^2 - tV + s = 0 \;\Rightarrow\; V = \frac{g}{2\ln 2}\left(t - \sqrt{t^2 - \tfrac{4\ln 2}{g}s}\right).$$

(The other root, ≈ 5100 ft/s, would give $gt/V \ll 1$, contradicting the assumption.) Plugging in $s = 29{,}300$ ft, $t = 116$ s, $g = 32.2$:

$$V \approx \frac{32.2}{1.386}\left(116 - \sqrt{13456 - 2523}\right) \approx \mathbf{266\ ft/s}\ (\approx 181\text{ mph}).$$

Solving the exact equation $s = \frac{V^2}{g}\ln\cosh(gt/V)$ numerically gives $V = 265.69$ ft/s, identical to the digits shown, since the approximation is excellent here.

Drag constant. From $V^2 = mg/k$ we get $k = mg/V^2 = W/V^2$, where $W = mg = 261.2$ lb is the weight (the mass is $m = W/g \approx 8.11$ slug):

$$\boxed{k = \frac{261.2}{(265.7)^2} \approx 3.7\times 10^{-3}\ \text{lb}\cdot\text{s}^2/\text{ft}^2} \qquad (k/m \approx 4.6\times 10^{-4}\ \text{ft}^{-1}).$$

■

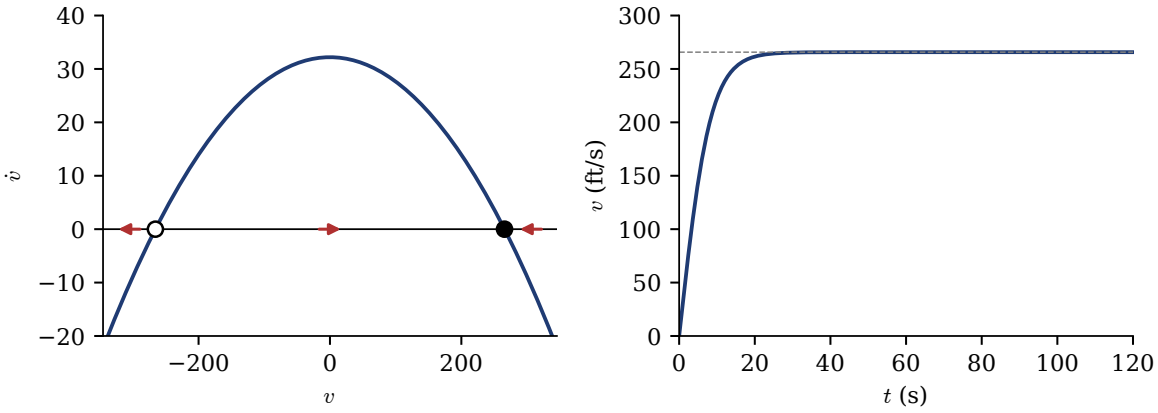


Figure 12: 2.2.13: phase line of $\dot v = g(1 - v^2/V^2)$ and $v(t) = V\tanh(gt/V)$ for $V = 265.7$ ft/s.

2.3 Population Growth

2.3.1 *(Logistic equation) Solve $\dot N = rN(1 - N/K)$ for arbitrary N_0 (a) by partial fractions, (b) with $x = 1/N$.*

Solution. **(a)** Since $\dfrac{1}{N(1-N/K)} = \dfrac{1}{N} + \dfrac{1}{K-N}$,

$$\int \Big(\frac{1}{N} + \frac{1}{K-N}\Big)\mathrm{d}N = \ln\left|\frac{N}{K-N}\right| = rt + c \;\Rightarrow\; \frac{N}{K-N} = \frac{N_0}{K-N_0}\mathrm{e}^{rt}.$$

Solving for N:

$$\boxed{N(t) = \frac{K}{1 + \big(\frac{K}{N_0} - 1\big)\mathrm{e}^{-rt}}}.$$

(b) With $x = 1/N$: $\dot{x} = -\dot{N}/N^2 = -\dfrac{r}{N}\Big(1 - \dfrac{N}{K}\Big) = -rx + \dfrac{r}{K}$, a linear equation with solution $x(t) = \dfrac{1}{K} + \Big(\dfrac{1}{N_0} - \dfrac{1}{K}\Big)\mathrm{e}^{-rt}$. Inverting, $N = 1/x$ gives the same formula as in (a). ■

2.3.2 *(Autocatalysis)* $\dot{x} = k_1 a x - k_{-1}x^2$, $k_1, k_{-1} > 0$. *(a) Find and classify the fixed points. (b) Sketch* $x(t)$.

Solution. **(a)** $\dot{x} = x(k_1 a - k_{-1}x)$, so $x^* = 0$ and $x^* = k_1 a/k_{-1}$. With $f'(x) = k_1 a - 2k_{-1}x$: $f'(0) = k_1 a > 0$ (**unstable**) and $f'(k_1 a/k_{-1}) = -k_1 a < 0$ (**stable**). So an infinitesimal amount of X catalyzes its own production up to the equilibrium concentration $k_1 a/k_{-1}$, where the forward and back reactions balance.

(b) The equation is a logistic equation with $r = k_1 a$, $K = k_1 a/k_{-1}$: for $x_0 > 0$, $x(t) = K/\big[1 + (K/x_0 - 1)\mathrm{e}^{-k_1 a t}\big]$. Solutions from $0 < x_0 < K/2$ are sigmoidal (S-shaped), those from $x_0 > K$ decay monotonically to K, and $x_0 = 0$ stays at 0 (Fig. 13; $x_0 < 0$ is unphysical). ■

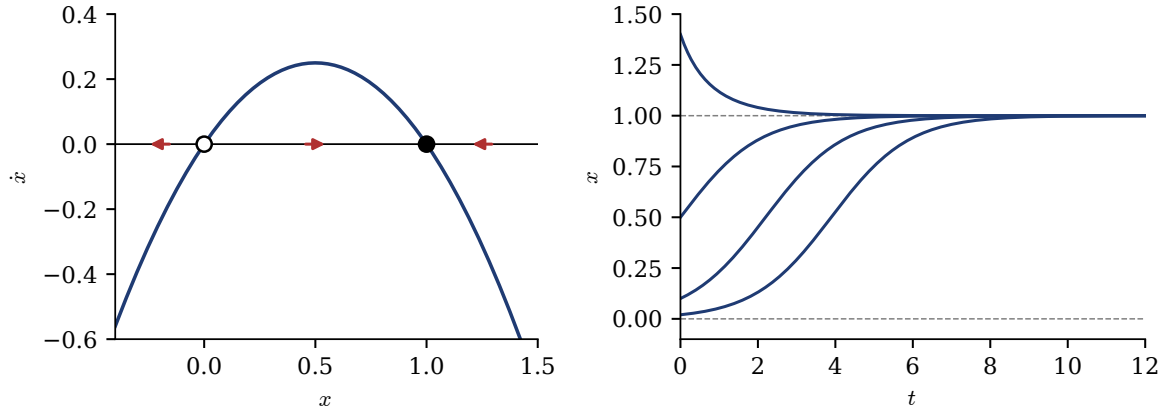


Figure 13: 2.3.2 with $k_1 a = k_{-1} = 1$.

2.3.3 *(Tumor growth) Gompertz law* $\dot{N} = -aN\ln(bN)$, $a, b > 0$. *(a) Interpret* a, b. *(b) Sketch the vector field and* $N(t)$.

Solution. **(a)** Write $\dot{N}/N = a\ln\big(\frac{1}{bN}\big)$. The per-capita growth rate is positive for $N < 1/b$, zero at $N = 1/b$ and negative for $N > 1/b$. Thus $1/b$ is the *carrying capacity* (the maximal sustainable tumor size, limited by nutrient and oxygen supply), and a sets the *time scale* (the growth rate):

the per-capita rate has slope $-a$ per unit of $\ln N$. Unlike the logistic law, the per-capita rate is unbounded as $N \to 0$.

(b) $f(N) = -aN\ln(bN)$ is positive on $(0, 1/b)$ and negative for $N > 1/b$, with $f \to 0$ as $N \to 0^+$. Hence $N = 1/b$ is stable and small tumors grow away from $N = 0$. Exact solution: with $y = \ln(bN)$, $\dot{y} = \dot{N}/N = -ay$, so $y = y_0 \mathrm{e}^{-at}$ and

$$N(t) = \frac{1}{b}\,(bN_0)^{\mathrm{e}^{-at}}.$$

$N(t)$ is S-shaped, with inflection where $\ddot{N} = f'(N)\dot{N} = 0$, i.e. at $N = 1/(\mathrm{e}b)$, and tends to $1/b$ (Fig. 14). ■

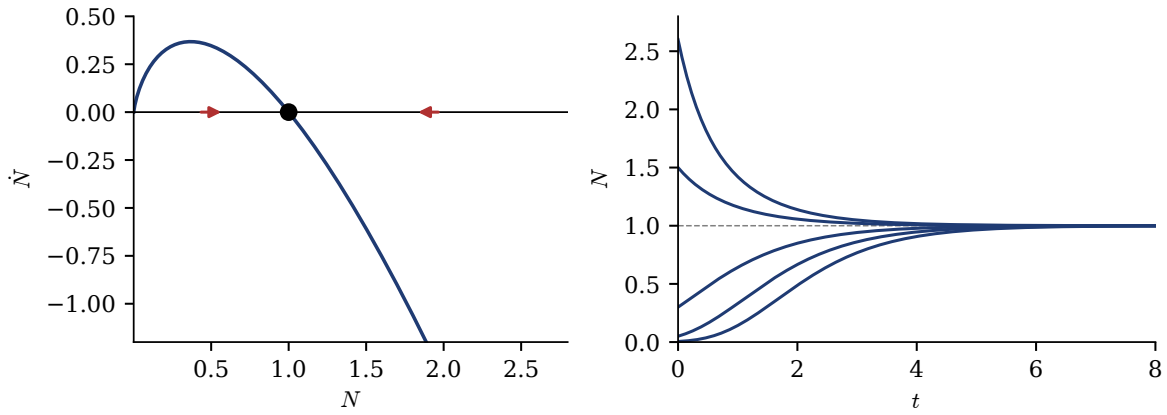


Figure 14: 2.3.3 with $a = b = 1$ (the open dot at $N = 0$ is a limiting fixed point).

2.3.4 *(The Allee effect)* $\dot{N}/N = r - a(N-b)^2$. *(a) Show this is an Allee effect for suitable constants. (b) Fixed points and stability. (c) Sketch* $N(t)$. *(d) Compare with the logistic equation.*

Solution. (a) The per-capita rate $r - a(N-b)^2$ is a downward parabola with its maximum r at $N = b$. It is highest at an intermediate density (the Allee effect) when $a > 0$, $b > 0$, $r > 0$. For the *strong* version (growth is negative at very low density because mates are hard to find) we also need the rate to be negative at $N = 0$: $r - ab^2 < 0$, i.e. $ab^2 > r$. Assume this from now on.

(b) $\dot{N} = N\big[r - a(N-b)^2\big]$ has fixed points $N^* = 0$ and $N_\pm = b \pm \sqrt{r/a}$, both positive since $\sqrt{r/a} < b$. With $f'(N) = r - a(N-b)^2 - 2aN(N-b)$:

- $f'(0) = r - ab^2 < 0$: $N^* = 0$ **stable** (extinction);
- at $N_\pm$, $(N-b)^2 = r/a$ so $f'(N_\pm) = -2aN_\pm(N_\pm - b) = \mp 2aN_\pm\sqrt{r/a}$: N_- is **unstable** (the extinction threshold) and N_+ is **stable** (the carrying capacity).

(c) If $N_0 < N_-$ the population dies out; if $N_- < N_0 < N_+$ it grows to N_+; if $N_0 > N_+$ it decreases to N_+ (Fig. 15).

(d) The logistic equation has a single stable positive equilibrium and $N = 0$ is unstable, so *any* small population grows. With the Allee effect the system is *bistable*: there is a critical population size N_- below which the species goes extinct (a threshold that logistic growth lacks). (If $r > ab^2$ the Allee effect is only "weak": $N = 0$ is unstable, $N_- < 0$, and the dynamics are qualitatively logistic. If $r < 0$ every population goes extinct.) ■

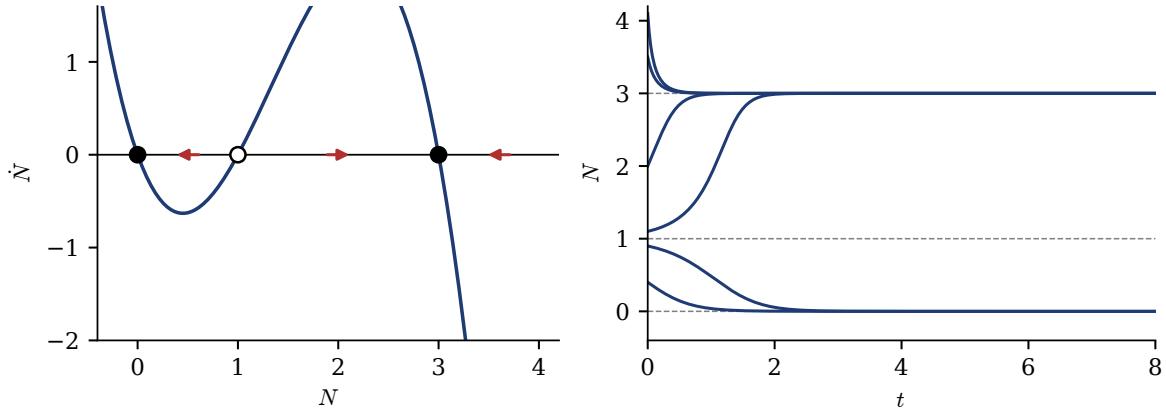


Figure 15: 2.3.4 with $r = a = 1$, $b = 2$: $N_- = 1$, $N_+ = 3$.

2.3.5 *(Dominance of the fittest)* $\dot{X} = aX$, $\dot{Y} = bY$, $a > b > 0$, $X_0, Y_0 > 0$; $x = X/(X+Y)$. *(a) Show* $x \uparrow 1$ *using the explicit solutions. (b) Show* $\dot{x} = (a-b)x(1-x)$.

Solution. **(a)** $X = X_0 e^{at}$, $Y = Y_0 e^{bt}$, so

$$x(t) = \frac{X_0 e^{at}}{X_0 e^{at} + Y_0 e^{bt}} = \frac{1}{1 + \frac{Y_0}{X_0} e^{-(a-b)t}}.$$

Since $a - b > 0$ the denominator decreases strictly from $1 + Y_0/X_0$ to 1, so x increases monotonically and $x \to 1$ as $t \to \infty$.

(b) By the quotient rule,

$$\dot{x} = \frac{\dot{X}(X+Y) - X(\dot{X}+\dot{Y})}{(X+Y)^2} = \frac{aX(X+Y) - X(aX+bY)}{(X+Y)^2} = (a-b)\frac{XY}{(X+Y)^2} = (a-b)\,x(1-x).$$

This is the logistic equation with $r = a - b > 0$ and $K = 1$. Its fixed points are $x = 0$ (unstable) and $x = 1$ (stable), and a solution in $(0,1)$ cannot cross them, so it increases monotonically and tends to 1 (a 1D flow cannot oscillate or turn around). ■

2.3.6 *(Language death)* $\dot{x} = s(1-x)x^a - (1-s)x(1-x)^a$ *with* $0 < s < 1$ *and* $a > 1$. *(a) Three fixed points. (b)* $x = 0, 1$ *stable. (c) The third is unstable.*

Solution. Factor: $f(x) = x(1-x)\,h(x)$ with

$$h(x) = s\,x^{a-1} - (1-s)(1-x)^{a-1}.$$

(a) $f(0) = f(1) = 0$. In $(0,1)$ the zeros of f are the zeros of h. Since $a > 1$, x^{a-1} is strictly increasing and $(1-x)^{a-1}$ strictly decreasing, so h is strictly increasing, with $h(0) = -(1-s) < 0$ and $h(1) = s > 0$. By the intermediate value theorem and monotonicity, h has *exactly one* zero $x^* \in (0,1)$. Hence f has exactly three fixed points: 0, x^*, 1. (For $a = 2$: $x^* = 1 - s$.)

(b) Differentiate: $f'(x) = -s\,x^a + a\,s(1-x)x^{a-1} - (1-s)(1-x)^a + (1-s)\,a\,x(1-x)^{a-1}$. Because $a > 1$ the terms with exponent $a - 1 > 0$ vanish at $x = 0$ or $x = 1$, giving

$$f'(0) = -(1-s) < 0, \qquad f'(1) = -s < 0,$$

so both $x = 0$ and $x = 1$ are **stable**.

(c) At x^*, $h(x^*) = 0$, so $f'(x^*) = x^*(1 - x^*)\,h'(x^*)$ and $h'(x^*) > 0$ (as h is strictly increasing: $h'(x) = (a-1)\big[s\,x^{a-2} + (1-s)(1-x)^{a-2}\big] > 0$). So $f'(x^*) > 0$: x^* is **unstable**.

Interpretation: two languages cannot coexist stably; the language whose speakers start above the threshold x^* takes over (Fig. 16). ■

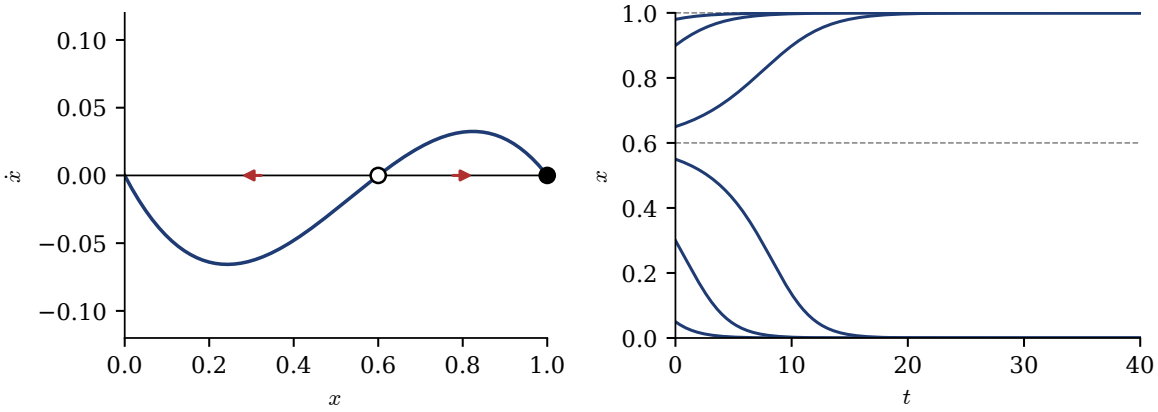


Figure 16: 2.3.6 with $a = 2$, $s = 0.4$ (so $x^* = 0.6$).

2.4 Linear Stability Analysis

2.4.1–2.4.6 *Classify the fixed points of each system using $f'(x^*)$; if $f'(x^*) = 0$ use a graphical argument.*

Solution.

2.4.1 $\dot{x} = x(1-x)$**.** $f'(x) = 1 - 2x$. $x^* = 0$: $f' = 1 > 0$, **unstable**; $x^* = 1$: $f' = -1 < 0$, **stable**.

2.4.2 $\dot{x} = x(1-x)(2-x)$**.** Fixed points $0, 1, 2$. Expanding, $f = 2x - 3x^2 + x^3$, $f' = 2 - 6x + 3x^2$. $f'(0) = 2 > 0$: **unstable**; $f'(1) = -1 < 0$: **stable**; $f'(2) = 2 > 0$: **unstable**.

2.4.3 $\dot{x} = \tan x$**.** Fixed points $x^* = n\pi$ ($\tan x$ is undefined at $\frac{\pi}{2} + n\pi$, where the velocity is infinite; the flow lives on each interval $(n\pi - \frac{\pi}{2}, n\pi + \frac{\pi}{2})$). $f'(x) = \sec^2 x$, so $f'(n\pi) = 1 > 0$: all fixed points are **unstable**.

2.4.4 $\dot{x} = x^2(6-x)$**.** Fixed points $x^* = 0$ (double root) and $x^* = 6$. $f'(x) = 12x - 3x^2$: $f'(6) = -36 < 0$, **stable**. $f'(0) = 0$: linearization fails. But $f(x) = x^2(6-x) > 0$ for all $x < 6$, $x \neq 0$, so the flow is to the right on both sides of 0: $x^* = 0$ is **half-stable** (attracting from the left, repelling to the right).

2.4.5 $\dot{x} = 1 - \mathrm{e}^{-x^2}$**.** $f = 0 \iff x = 0$. $f'(x) = 2x\mathrm{e}^{-x^2}$, $f'(0) = 0$: inconclusive. Since $\mathrm{e}^{-x^2} < 1$ for $x \neq 0$, $f > 0$ on both sides, so the flow is to the right on both sides: $x^* = 0$ is **half-stable**.

2.4.6 $\dot{x} = \ln x$ $(x > 0)$. $x^* = 1$; $f'(x) = 1/x$, $f'(1) = 1 > 0$: **unstable**. (For $0 < x < 1$, $\dot{x} < 0$ and the solution reaches $x = 0$ in finite time, where $\ln x$ is singular.)

■

2.4.7 *$\dot{x} = ax - x^3$ with $a > 0$, $a < 0$ and $a = 0$.*

Solution. $f'(x) = a - 3x^2$ and $f(x) = x(a - x^2)$.

- $a > 0$: fixed points $0, \pm\sqrt{a}$. $f'(0) = a > 0$: **unstable**; $f'(\pm\sqrt{a}) = a - 3a = -2a < 0$: **stable**.
- $a < 0$: the only fixed point is $x^* = 0$ ($a - x^2 < 0$ has no real roots), and $f'(0) = a < 0$: **stable**.
- $a = 0$: $\dot{x} = -x^3$, $f'(0) = 0$, linearization fails. Graphically $f > 0$ for $x < 0$ and $f < 0$ for $x > 0$, so $x^* = 0$ is **stable** (though the approach is algebraic, not exponential; see 2.4.9).

Passing from $a < 0$ to $a > 0$ the stable origin gives birth to two new stable fixed points $\pm\sqrt{a}$ while itself becoming unstable: a *supercritical pitchfork bifurcation* (Chapter 3). ■

2.4.8 *Classify the fixed points of the Gompertz model $\dot{N} = -aN\ln(bN)$ by linear stability.*

Solution. $f'(N) = -a\ln(bN) - a$. The only positive fixed point is $N^* = 1/b$ with $f'(1/b) = -a < 0$: **stable**. At $N = 0$: $f(N) \to 0$ as $N \to 0^+$, so $N = 0$ is a limiting fixed point, but $f'(N) = -a[\ln(bN) + 1] \to +\infty$ as $N \to 0^+$ and $f > 0$ for small N, so it is **unstable** (and linearization is not meaningful there, since f is not differentiable at 0). ■

2.4.9 *(Critical slowing down) Consider $\dot{x} = -x^3$. (a) Find $x(t)$; show $x \to 0$ but not exponentially. (b) Plot the solution for $x_0 = 10$, $0 \le t \le 10$, together with that of $\dot{x} = -x$.*

Solution. **(a)** Separating variables, $\int x^{-3}\mathrm{d}x = -\int \mathrm{d}t$ gives $-\frac{1}{2x^2} = -t + c$, i.e. $\frac{1}{x^2} = \frac{1}{x_0^2} + 2t$, so

$$\boxed{x(t) = \frac{x_0}{\sqrt{1 + 2x_0^2 t}}}.$$

As $t \to \infty$, $x(t) \sim \mathrm{sgn}(x_0)/\sqrt{2t} \to 0$, but only *algebraically*. It is not exponential: for any $\lambda > 0$, $|x(t)|\,\mathrm{e}^{\lambda t} \sim \mathrm{e}^{\lambda t}/\sqrt{2t} \to \infty$, so $x(t)$ decays more slowly than every exponential. This is critical slowing down: the linearization $\dot{x} \approx f'(0)x = 0$ at the fixed point has zero rate, so the relaxation time is infinite.

(b) For $x_0 = 10$: $x(t) = 10/\sqrt{1 + 200t}$ versus $x(t) = 10\mathrm{e}^{-t}$. At $t = 10$ these are ≈ 0.22 and $\approx 4.5 \times 10^{-4}$ (Fig. 17). The semilog plot shows the exponential as a straight line, whereas the cubic case curves, with the slope tending to 0. ■

2.5 Existence and Uniqueness

2.5.1 *(Reaching a fixed point in finite time) A particle moves on $x \ge 0$ with $\dot{x} = -x^c$. (a) For which c is the origin a stable fixed point? (b) If so, how long does it take to go from $x = 1$ to $x = 0$?*

Solution. **(a)** $x = 0$ is a fixed point only if $f(0) = 0$, i.e. $c > 0$ (for $c \le 0$, x^c is infinite or equal to 1 at 0). For $c > 0$, $f(x) = -x^c < 0$ on $x > 0$, so the flow is toward 0: the origin is stable for **all** $c > 0$.

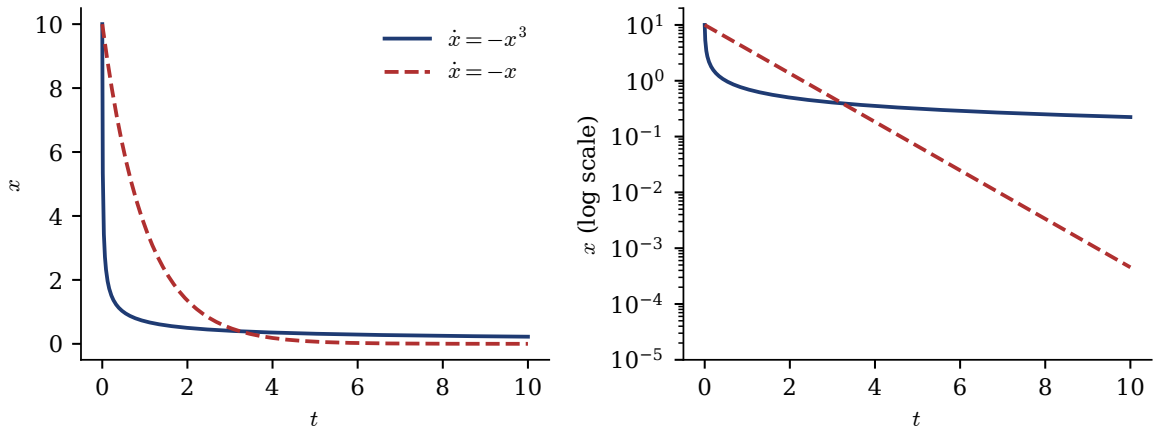


Figure 17: 2.4.9: algebraic decay (solid) vs. exponential decay (dashed), linear and semilog axes.

(b) Since $\mathrm{d}t = -\mathrm{d}x/x^c$, the time from $x = 1$ to 0 is

$$T = \int_0^1 x^{-c}\,\mathrm{d}x = \left[\frac{x^{1-c}}{1-c}\right]_0^1 = \frac{1}{1-c} \quad \text{if } 0 < c < 1,$$

while the integral diverges if $c \geq 1$. So the origin is reached in **finite time iff** $0 < c < 1$, with $T = \dfrac{1}{1-c}$ (which $\to \infty$ as $c \to 1^-$; for $c = 1$ the approach is exponential, for $c > 1$ algebraic). Note that f is not Lipschitz at 0 when $c < 1$, which is exactly why the existence-and-uniqueness theorem does not rule this out. ■

2.5.2 *(Blow-up) Show that the solution of $\dot{x} = 1 + x^{10}$ escapes to $+\infty$ in finite time from any initial condition.*

Solution. Since $\dot{x} \geq 1 > 0$, $x(t)$ is strictly increasing and $x(t) \geq x_0 + t$, so it is unbounded. The time needed to reach a level X is

$$t(X) = \int_{x_0}^{X} \frac{\mathrm{d}x}{1 + x^{10}} \;\leq\; \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1 + x^{10}} = \frac{\pi/5}{\sin(\pi/10)} \approx 2.03,$$

which is bounded *independently of X and x_0*. Hence $x(t) \to +\infty$ at some finite time $t^* \leq 2.03$.

(*Following the hint:* for $x \geq 1$, $1 + x^{10} \geq 1 + x^2$, so the solution is at least as fast as that of $\dot{y} = 1 + y^2$, namely $y = \tan(t + c)$, which blows up in finite time; by comparison x blows up no later.) ■

2.5.3 *Show that $x(t) \to \pm\infty$ in finite time for $\dot{x} = rx + x^3$, $r > 0$, from any $x_0 \neq 0$.*

Solution. $f(x) = x(r + x^2)$ vanishes only at $x = 0$; f is odd, so take $x_0 > 0$ (the case $x_0 < 0$ is symmetric). Then $\dot{x} > 0$ and the time to reach $+\infty$ is

$$T = \int_{x_0}^{\infty} \frac{\mathrm{d}x}{x(r + x^2)} = \frac{1}{r}\int_{x_0}^{\infty} \left(\frac{1}{x} - \frac{x}{r + x^2}\right)\mathrm{d}x = \frac{1}{2r}\left[\ln \frac{x^2}{r + x^2}\right]_{x_0}^{\infty} = \frac{1}{2r}\ln\left(1 + \frac{r}{x_0^2}\right) < \infty.$$

Equivalently, with $u = x^{-2}$, $\dot{u} = -2x^{-3}\dot{x} = -2ru - 2$, so $u = (u_0 + \frac{1}{r})\mathrm{e}^{-2rt} - \frac{1}{r}$, which hits $u = 0$ (i.e. $x = \infty$) at $t = T$ above. For $x_0 < 0$, $x \to -\infty$ at the same time T. ■

2.5.4 *(Infinitely many solutions with the same initial condition) Show that $\dot{x} = x^{1/3}$, $x(0) = 0$, has infinitely many solutions.*

Solution. $x(t) \equiv 0$ is one solution. For others separate variables for $x \neq 0$: $\int x^{-1/3}\mathrm{d}x = \frac{3}{2}x^{2/3} = t - t_0$, i.e. $x = \pm\big[\frac{2}{3}(t-t_0)\big]^{3/2}$ for $t \geq t_0$. For any $t_0 \geq 0$ define

$$x(t) = \begin{cases} 0, & 0 \leq t \leq t_0, \\ \pm\big[\frac{2}{3}(t-t_0)\big]^{3/2}, & t > t_0. \end{cases}$$

Check: for $t > t_0$, $\dot{x} = \pm\big[\frac{2}{3}(t-t_0)\big]^{1/2} = x^{1/3}$ (the cube root of $\pm[\cdot]^{3/2}$ is $\pm[\cdot]^{1/2}$). At $t = t_0$ the derivative from the right is 0, matching the left, so x is C^1 and solves the ODE everywhere. Each $t_0 \geq 0$ and each sign gives a different solution with $x(0) = 0$ (Fig. 18): the particle sits at 0 for an arbitrary time and then "takes off". This is possible because $f(x) = x^{1/3}$ has infinite slope at 0 (not Lipschitz). ■

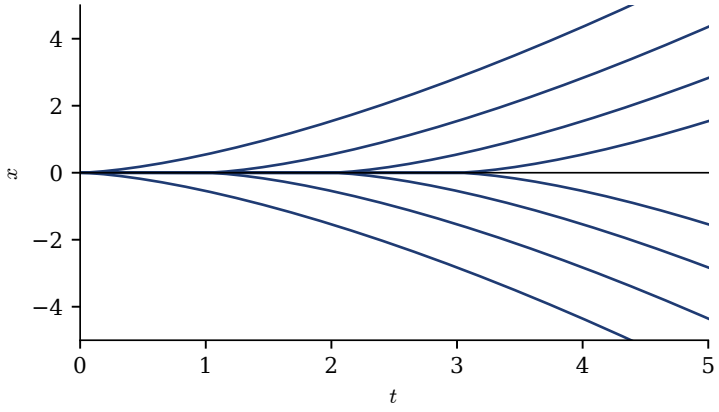


Figure 18: 2.5.4: solutions with $t_0 = 0, 1, 2, 3$ (both signs).

2.5.5 *(Non-uniqueness) Consider $\dot{x} = |x|^{p/q}$, $x(0) = 0$, with p, q positive coprime integers. (a) Show there are infinitely many solutions if $p < q$. (b) Show the solution is unique if $p > q$.*

Solution. Let $r = p/q$.

(a) $p < q \Rightarrow 0 < r < 1$. $x \equiv 0$ is a solution. For $x > 0$, $x^{-r}\mathrm{d}x = \mathrm{d}t$ gives $\frac{x^{1-r}}{1-r} = t - t_0$, so for any $t_0 \geq 0$

$$x(t) = \begin{cases} 0, & t \leq t_0, \\ \big[(1-r)(t-t_0)\big]^{1/(1-r)}, & t > t_0, \end{cases} \qquad \frac{1}{1-r} = \frac{q}{q-p} > 1,$$

is a solution (it is C^1 at t_0 because the exponent $\frac{1}{1-r} > 1$; and $\dot{x} = \big[(1-r)(t-t_0)\big]^{r/(1-r)} = x^r$). Infinitely many solutions.

(b) $p > q \Rightarrow r > 1$; then $f(x) = |x|^r$ is C^1 (its derivative $r|x|^{r-1}\operatorname{sgn} x$ is continuous), hence locally Lipschitz, and the existence–uniqueness theorem gives a unique solution, $x \equiv 0$. Direct proof: $\dot{x} \geq 0$, so x is nondecreasing and $x(t) \geq 0$. Suppose $x(t_1) > 0$ for some $t_1 > 0$ and let $t_* = \sup\{t < t_1 : x(t) = 0\}$; then $x > 0$ on $(t_*, t_1]$ and for small $\varepsilon > 0$

$$t_1 - t_* - \varepsilon = \int_{t_*+\varepsilon}^{t_1} \frac{\dot{x}}{x^r}\mathrm{d}t = \int_{x(t_*+\varepsilon)}^{x(t_1)} x^{-r}\mathrm{d}x \xrightarrow[\varepsilon\to 0]{} \int_0^{x(t_1)} x^{-r}\mathrm{d}x = \infty \quad (r \geq 1),$$

a contradiction since the left side is finite. (This also covers $p = q$.) ■

2.5.6 *(The leaky bucket) (a) Show $a\,v = A\,\dot{h}$ and name the physical law. (b) Derive $v^2 = 2gh$ by energy conservation. (c) Show $\dot{h} = -C\sqrt{h}$ with $C = \sqrt{2g}\,(a/A)$. (d) Show that the solution with $h(0) = 0$ is non-unique backward in time.*

Solution. **(a)** Water is incompressible, so the volume that leaves through the hole per unit time, $a\,v$, equals the rate at which the bucket's volume decreases, $A|\dot{h}|$. With h decreasing, $\dot{h} < 0$, so the exact statement is $A\dot{h} = -av$, i.e. $av = A|\dot{h}|$ (the book's relation is for magnitudes; the sign is consistent with $\dot{h} = -C\sqrt{h}$ below). The law invoked is *conservation of mass* (continuity).

(b) When the level drops by Δh, a layer of mass $\Delta m = \rho A\,\Delta h$ is effectively removed from height h and leaves at the hole (height ≈ 0). The potential energy of the system decreases by $\Delta m\,g\,h = \rho A\,\Delta h\,g\,h$. The escaping water has kinetic energy $\frac{1}{2}\Delta m\,v^2 = \frac{1}{2}\rho A\,\Delta h\,v^2$. Assuming all of the potential energy is converted to kinetic energy, $\frac{1}{2}\rho A\Delta h\,v^2 = \rho A\Delta h\,gh$, so $v^2 = 2gh$ (Torricelli's law).

(c) $\dot{h} = -\dfrac{a}{A}v = -\dfrac{a}{A}\sqrt{2gh} = -C\sqrt{h}$ with $C = \sqrt{2g}\,\dfrac{a}{A}$.

(d) $x(t) \equiv 0$ ($h \equiv 0$) is a solution. For $h > 0$: $h^{-1/2}\mathrm{d}h = -C\,\mathrm{d}t \Rightarrow 2\sqrt{h} = C(t_0 - t)$, so $h(t) = \frac{C^2}{4}(t_0 - t)^2$ for $t \le t_0$. For any $t_0 \le 0$ define

$$h(t) = \begin{cases} \frac{C^2}{4}(t_0 - t)^2, & t \le t_0, \\ 0, & t \ge t_0. \end{cases}$$

This is C^1 at t_0 and satisfies $\dot{h} = -C\sqrt{h}$, with $h(0) = 0$. So the bucket is empty at $t = 0$ in all these solutions, but was full (at some height) at the arbitrary earlier time t_0: knowing that the bucket is empty now does not tell us when it was full; the solution is non-unique *backward in time* ($t < 0$), exactly because $\sqrt{h}$ is not Lipschitz at $h = 0$ (Fig. 19). ■

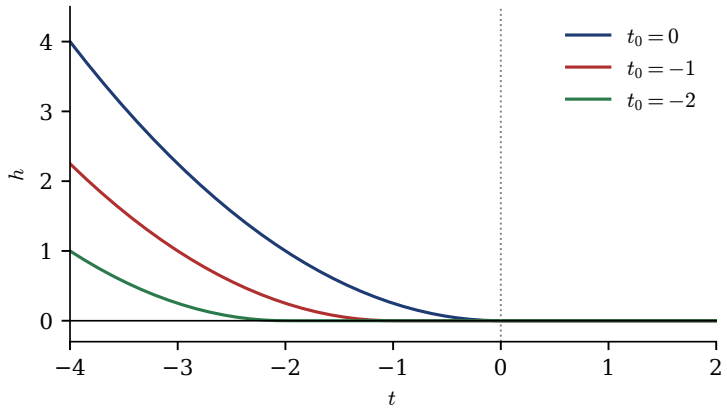


Figure 19: 2.5.6: $h(t)$ for $t_0 = 0, -1, -2$ (with $C = 1$).

2.6 Impossibility of Oscillations

2.6.1 *Explain the paradox: a harmonic oscillator $m\ddot{x} = -kx$ oscillates along a line, yet one-dimensional systems cannot oscillate.*

Solution. The dimension of a dynamical system is the number of independent state variables (the dimension of its *phase space*), not the dimension of the space in which the object moves. The equation $m\ddot{x} = -kx$ is second order: writing $v = \dot{x}$ it becomes $\dot{x} = v$, $\dot{v} = -kx/m$, a *two-dimensional* system with phase space the (x, v)-plane, where the trajectories are closed ellipses. The statement "no oscillations in 1D" refers to first-order systems $\dot{x} = f(x)$, in which the state is determined by x alone. ■

2.6.2 *(No periodic solutions to $\dot{x} = f(x)$) Suppose $x(t) = x(t+T)$ is a nontrivial periodic solution, with $x(t) \neq x(t+s)$ for $0 < s < T$. Derive a contradiction by considering $\int_t^{t+T} f(x)\frac{\mathrm{d}x}{\mathrm{d}t}\mathrm{d}t$.*

Solution. Compute the integral in two ways. First, substitute $u = x(t)$:

$$\int_t^{t+T} f(x)\frac{\mathrm{d}x}{\mathrm{d}t}\mathrm{d}t = \int_{x(t)}^{x(t+T)} f(u)\,\mathrm{d}u = 0,$$

because $x(t+T) = x(t)$ (the integral of a function between equal limits). Second, use the ODE $f(x) = \dot{x}$:

$$\int_t^{t+T} f(x)\,\dot{x}\,\mathrm{d}t = \int_t^{t+T} \dot{x}^{\,2}\,\mathrm{d}t > 0,$$

because $\dot{x}^2 \geq 0$ is continuous and not identically zero (a nontrivial periodic solution is not constant). So $0 > 0$, a contradiction. Hence no nonconstant periodic solutions exist. ■

2.7 Potentials

For $\dot{x} = f(x)$ the potential is $V(x) = -\int f(x)\,\mathrm{d}x$, so $f = -V'$; equilibria are the critical points of V, with minima stable and maxima unstable ($V'' = -f'$).

2.7.1–2.7.5 *Plot $V(x)$ and identify equilibria and stability for (1) $\dot{x} = x(1-x)$, (2) $\dot{x} = 3$, (3) $\dot{x} = \sin x$, (4) $\dot{x} = 2 + \sin x$, (5) $\dot{x} = -\sin x$.*

Solution.

2.7.1 $V = -\frac{x^2}{2} + \frac{x^3}{3}$ (up to a constant). $V' = -x + x^2 = 0 \Rightarrow x = 0, 1$; $V'' = -1 + 2x$: $V''(0) = -1 < 0$ (local max, **unstable**); $V''(1) = 1 > 0$ (local min, **stable**).

2.7.2 $V = -3x$, a straight line with slope -3: no equilibria (the particle always rolls downhill to the right).

2.7.3 $V = \cos x$; $V' = -\sin x$, $V'' = -\cos x$. Maxima at $x = 2n\pi$ (**unstable**), minima at $x = (2n+1)\pi$ (**stable**).

2.7.4 $V = -2x + \cos x$, $V' = -(2 + \sin x) \leq -1 < 0$: monotonically decreasing, no equilibria; a tilted washboard with wiggles but no local extrema.

2.7.5 $V = -\cos x$: minima at $x = 2n\pi$ (**stable**), maxima at $x = (2n+1)\pi$ (**unstable**), the reverse of 2.7.3.

■

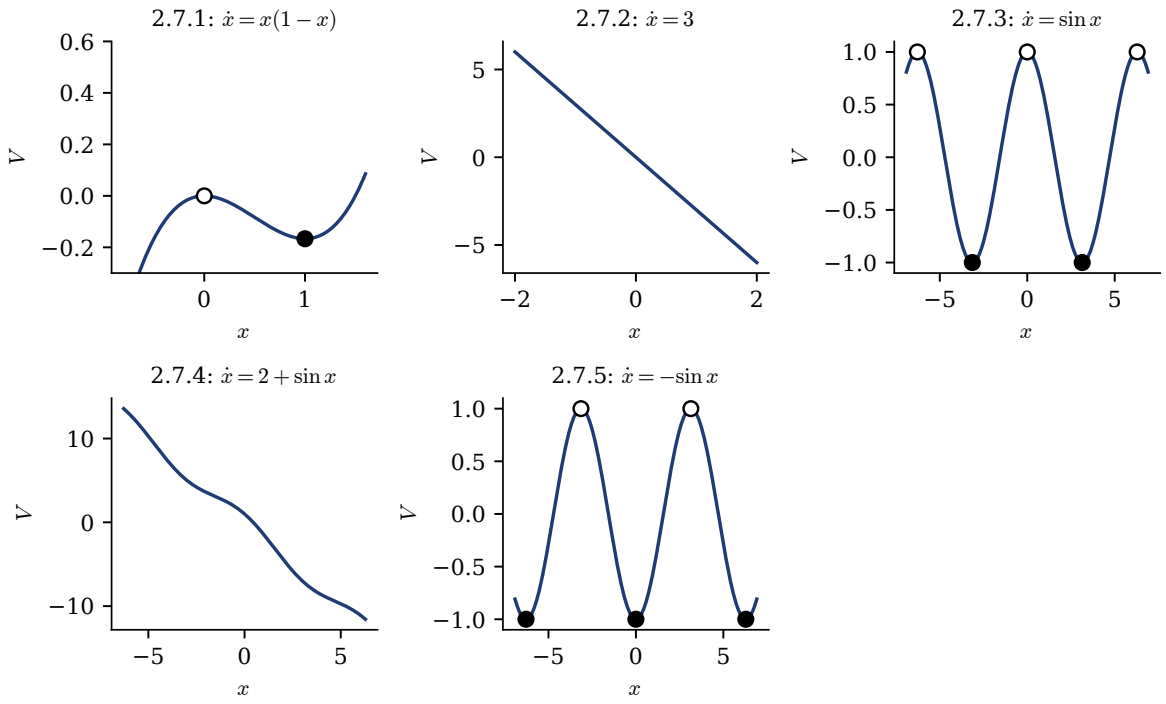


Figure 20: Potentials for 2.7.1–2.7.5 (filled: stable; open: unstable).

2.7.6 *$\dot{x} = r + x - x^3$ for various r.*

Solution. $V(x) = -rx - \frac{x^2}{2} + \frac{x^4}{4}$. Equilibria satisfy $x^3 - x = r$, the intersections of the cubic $y = x^3 - x$ with the horizontal line $y = r$. The cubic has a local max $y = \frac{2}{3\sqrt{3}}$ at $x = -\frac{1}{\sqrt{3}}$ and a local min $-\frac{2}{3\sqrt{3}}$ at $x = \frac{1}{\sqrt{3}}$. With $r_c = \frac{2}{3\sqrt{3}} \approx 0.385$:

- $|r| < r_c$: **three** equilibria $x_1 < -\frac{1}{\sqrt{3}} < x_2 < \frac{1}{\sqrt{3}} < x_3$ (double-well potential). Since $f' = 1 - 3x^2$ is negative for $|x| > 1/\sqrt{3}$ and positive for $|x| < 1/\sqrt{3}$: x_1, x_3 stable (minima), x_2 unstable (maximum). For $r = 0$: $x = 0$ (unstable), ± 1 (stable).
- $|r| = r_c$: **two** equilibria. For $r = r_c$, a double root at $x = -1/\sqrt{3}$ (half-stable, an inflection of V) and a simple stable root at $x = 2/\sqrt{3}$; for $r = -r_c$ the mirror image.
- $|r| > r_c$: only **one** equilibrium, stable (a single-well potential), at $x > 1$ if $r > r_c$ and $x < -1$ if $r < -r_c$.

The transitions at $r = \pm r_c$ are saddle-node bifurcations (Chapter 3). ■

2.7.7 *(Another proof that solutions cannot oscillate) Use the potential $V(x)$ to show that solutions of $\dot{x} = f(x)$ cannot oscillate.*

Solution. Let $V(x) = -\int^x f(u)\,\mathrm{d}u$, so $V' = -f$. Along any solution,

$$\frac{\mathrm{d}V}{\mathrm{d}t} = V'(x)\,\dot{x} = -f(x)\,\dot{x} = -\dot{x}^2 \le 0.$$

So $V(x(t))$ is nonincreasing, and strictly decreasing whenever $\dot{x} \neq 0$. If $x(t)$ were periodic with period T, then $V(x(t+T)) = V(x(t))$, so $0 = \Delta V = -\int_t^{t+T} \dot{x}^2\,\mathrm{d}t < 0$ unless $\dot{x} \equiv 0$, i.e. unless the

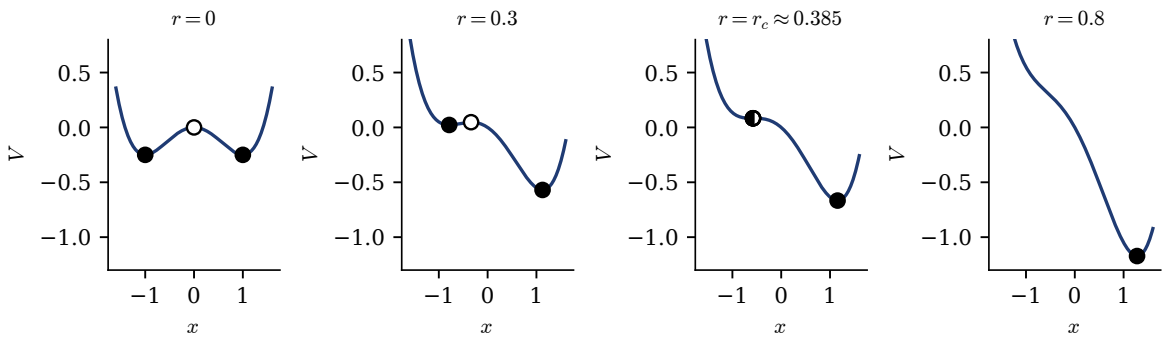


Figure 21: 2.7.6: $V(x)$ for $r = 0$, 0.3, r_c, 0.8 (half-filled dot: half-stable).

solution is constant. Contradiction: no nontrivial oscillations. (Physically: a particle in a viscous medium always loses "height" and never returns to its starting point.) ■

2.8 Solving Equations on the Computer

2.8.1 *Why is the slope constant along horizontal lines in the slope field of Fig. 2.8.2 ($\dot{x} = \sin x$)?*

Solution. In the (t, x)-plane the slope of a solution at (t, x) is $\mathrm{d}x/\mathrm{d}t = f(x) = \sin x$, which depends only on x, not on t (the equation is *autonomous*). Hence all points on a horizontal line $x = \text{const}$ have the same slope. Equivalently, a time-translate of a solution is again a solution. ■

2.8.2 *Sketch the slope field and integrate by hand: (a) $\dot{x} = x$; (b) $\dot{x} = 1 - x^2$; (c) $\dot{x} = 1 - 4x(1 - x)$; (d) $\dot{x} = \sin x$.*

Solution. In each case the slope is constant along horizontal lines (2.8.1), vanishes on the lines $x = x^*$, and one draws curves tangent to the field (Fig. 22).

(a) $\dot{x} = x$: slope $= x$; the horizontal line $x = 0$ is a solution; curves peel away exponentially ($x = x_0 \mathrm{e}^t$), growing for $x > 0$ and decreasing for $x < 0$.

(b) $\dot{x} = 1 - x^2$: horizontal solutions at $x = \pm 1$; slope is positive for $|x| < 1$, negative otherwise. All curves between and above converge to $x = 1$ (stable); those below $x = -1$ plunge to $-\infty$ (unstable at -1).

(c) $\dot{x} = 1 - 4x(1 - x) = (1 - 2x)^2 \geq 0$: slope vanishes only along $x = \frac{1}{2}$, otherwise positive. Curves are increasing; those with $x_0 < \frac{1}{2}$ approach $\frac{1}{2}$ from below, those with $x_0 > \frac{1}{2}$ leave it and blow up in finite time (half-stable fixed point).

(d) $\dot{x} = \sin x$: horizontal solutions at $x = n\pi$; curves are attracted to the odd multiples of π and repelled from the even ones.

■

2.8.3 *(Calibrating the Euler method) $\dot{x} = -x$, $x(0) = 1$. (a) Exact $x(1)$. (b) Euler estimates $\hat{x}(1)$ for $\Delta t = 10^{-n}$, $n = 0, \ldots, 4$. (c) Plot $E = |\hat{x}(1) - x(1)|$ vs. Δt, and $\ln E$ vs. $\ln \Delta t$.*

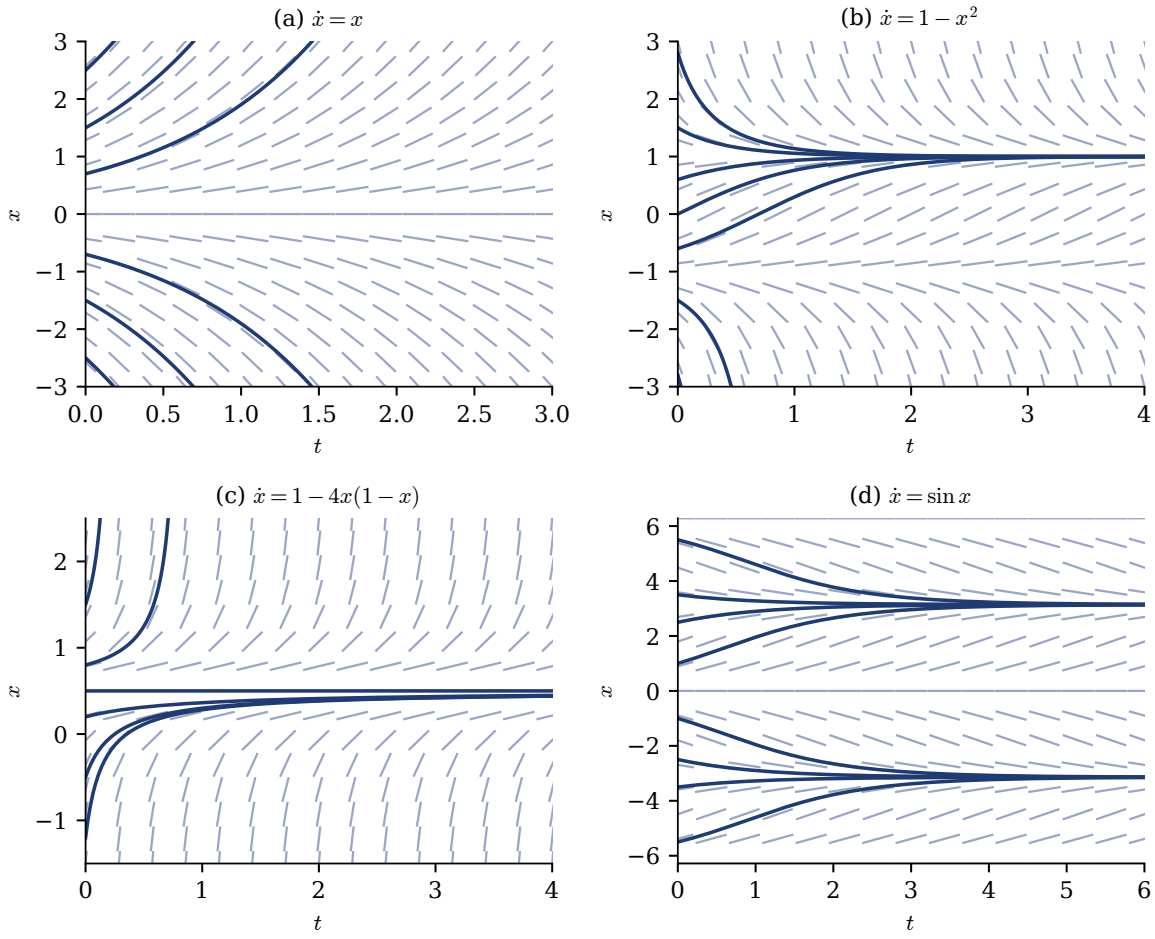


Figure 22: Slope fields (grey segments) and solution curves for 2.8.2.

Solution. **(a)** $x(t) = \mathrm{e}^{-t}$, so $x(1) = \mathrm{e}^{-1} = 0.3678794412\ldots$

(b) The Euler step is $x_{k+1} = x_k + \Delta t\,(-x_k) = (1-\Delta t)\,x_k$, so after $N = 1/\Delta t$ steps $\hat{x}(1) = (1-\Delta t)^{1/\Delta t}$. Results are in Table 1 (column "Euler").

(c) Expanding, $\hat{x}(1) = \exp\big[\frac{1}{\Delta t}\ln(1-\Delta t)\big] = \exp\big[-1 - \frac{\Delta t}{2} - \frac{\Delta t^2}{3} - \ldots\big] = \mathrm{e}^{-1}\big(1 - \frac{\Delta t}{2} + O(\Delta t^2)\big)$, so

$$E \approx \frac{\mathrm{e}^{-1}}{2}\,\Delta t \approx 0.184\,\Delta t.$$

Thus E is proportional to Δt (first order), and on a log–log plot $\ln E = \ln(0.184) + 1 \cdot \ln \Delta t$ is a straight line of **slope** 1. The measured slopes between successive points are 1.017, 1.002, 1.0002. Dividing the step by 10 reduces the error by 10, i.e. one additional correct digit per factor of 10 in work. (Fig. 23.) ■

2.8.4 *Redo 2.8.3 using the improved Euler method.*

Solution. The improved Euler (Heun) step is $\tilde{x} = x_k + \Delta t\, f(x_k)$, $x_{k+1} = x_k + \frac{1}{2}\Delta t\,[f(x_k) + f(\tilde{x})]$. For $f = -x$: $\tilde{x} = (1-\Delta t)x_k$ and $x_{k+1} = x_k\big[1 - \frac{1}{2}\Delta t(1 + (1-\Delta t))\big] = \big(1 - \Delta t + \frac{1}{2}\Delta t^2\big)x_k$.

This is the Taylor polynomial of $e^{-\Delta t}$ through second order, which explains the extra accuracy: $1-\Delta t+\frac{\Delta t^2}{2}=e^{-\Delta t}\big(1-\frac{\Delta t^3}{6}+\dots\big)$, so $\hat{x}(1)=e^{-1}\big(1-\frac{\Delta t^2}{6}+\dots\big)$ and

$$E\approx\frac{e^{-1}}{6}\Delta t^2\approx 0.0613\,\Delta t^2.$$

The log–log slope is **2** (measured: 2.03, 2.003, 2.0003); see Table 1. Reducing Δt by 10 reduces the error by 100. ■

2.8.5 *Redo 2.8.3 using the Runge–Kutta method.*

Solution. For $f=-x$ the four stages give $k_1=-x$, $k_2=-x(1-\frac{\Delta t}{2})$, $k_3=-x(1-\frac{\Delta t}{2}+\frac{\Delta t^2}{4})$, $k_4=-x(1-\Delta t+\frac{\Delta t^2}{2}-\frac{\Delta t^3}{4})$, so

$$x_{k+1}=\Big(1-\Delta t+\frac{\Delta t^2}{2}-\frac{\Delta t^3}{6}+\frac{\Delta t^4}{24}\Big)x_k,$$

the Taylor polynomial of $e^{-\Delta t}$ through fourth order $=e^{-\Delta t}\big(1-\frac{\Delta t^5}{120}+\dots\big)$. Hence $\hat{x}(1)=e^{-1}\big(1-\frac{\Delta t^4}{120}+\dots\big)$ and

$$E\approx\frac{e^{-1}}{120}\Delta t^4\approx 3.07\times10^{-3}\,\Delta t^4,$$

with log–log slope **4** (measured 4.03, 4.003, 4.0003). Already for $\Delta t=1$, $\hat{x}(1)=0.375$ ($E=7\times10^{-3}$). ■

Table 1: Results for 2.8.3–2.8.5: $\dot{x}=-x$, $x(0)=1$, exact $x(1)=0.36787944$. (Computed in 60-digit arithmetic to avoid round-off for the tiny RK errors.)

	Euler		Improved Euler		Runge–Kutta	
Δt	$\hat{x}(1)$	E	$\hat{x}(1)$	E	$\hat{x}(1)$	E
10^{0}	0	3.68×10^{-1}	0.5	1.32×10^{-1}	0.375	7.12×10^{-3}
10^{-1}	0.34868	1.92×10^{-2}	0.368541	6.62×10^{-4}	0.36787977	3.33×10^{-7}
10^{-2}	0.366032	1.85×10^{-3}	0.36788562	6.18×10^{-6}	0.3678794412	3.09×10^{-11}
10^{-3}	0.367695	1.84×10^{-4}	0.36787950	6.14×10^{-8}	0.3678794412	3.07×10^{-15}
10^{-4}	0.367861	1.84×10^{-5}	0.36787944	6.13×10^{-10}	0.3678794412	3.07×10^{-19}

2.8.6 *(Analytically intractable problem) $\dot{x}=x+e^{-x}$, $x(0)=0$. (a) Sketch $x(t)$ for $t\ge0$. (b) Prove $a<x(1)<b$ for a,b as close as you can. (c) Use Euler to compute $x(1)$ to three decimals; how small must Δt be? (d) Repeat (b)/(c) with Runge–Kutta.*

Solution. **(a)** $f(x)=x+e^{-x}$ has $f'=1-e^{-x}$, so its minimum is $f(0)=1$; thus $f\ge1>0$ for all x, there are no fixed points, and $x(t)$ increases monotonically with $\dot{x}\ge1$ ($x(t)\ge t$). Initially $\dot{x}(0)=1$, $\ddot{x}=f'(x)\dot{x}=(1-e^{-x})\dot{x}\ge0$, so the curve starts with slope 1 and is convex, and since $f(x)\sim x$ for large x it eventually grows exponentially. (Fig. 24.)

(b) *Lower bound.* $e^{-x}\ge1-x$ gives $\dot{x}\ge1$, so $x(1)\ge1$. A sharper lower bound is obtained below.

Upper bound. For $x\ge0$, $e^{-x}\le1-x+\frac{x^2}{2}$, so $\dot{x}\le1+\frac{x^2}{2}$. Compare with $\dot{y}=1+\frac{1}{2}y^2$, $y(0)=0$, whose solution is $y=\sqrt{2}\tan(t/\sqrt{2})$; since $x\le y$ (one-dimensional comparison principle),

$$x(1)\le\sqrt{2}\tan\frac{1}{\sqrt{2}}\approx1.2085=:b.$$

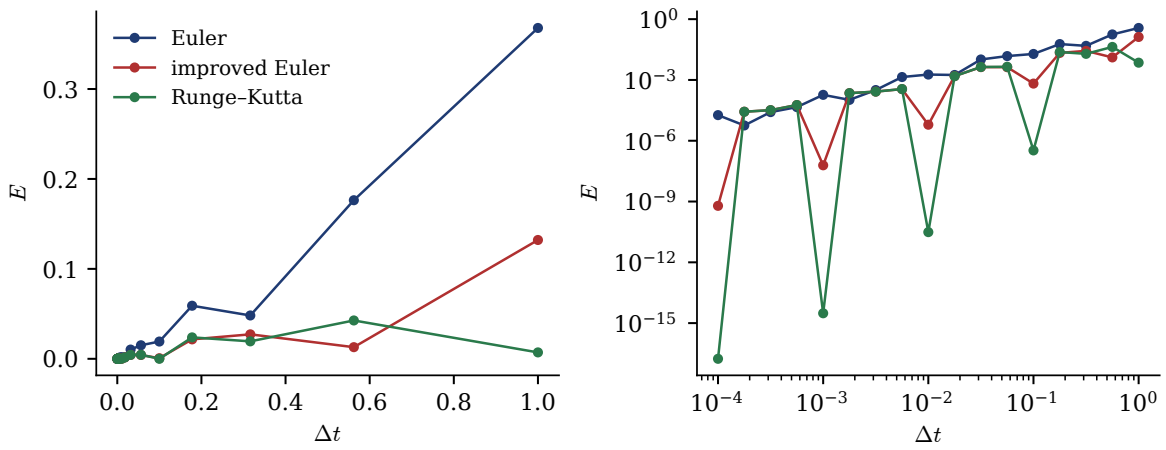


Figure 23: Error E vs. Δt (left, linear axes; right, log–log): slopes 1, 2, 4 for Euler, improved Euler, Runge–Kutta.

Sharper lower bound. For $x \geq 0$, $\mathrm{e}^{-x} \geq 1 - x + \frac{x^2}{2} - \frac{x^3}{6}$ (alternating Taylor series), so $\dot{x} \geq 1 + \frac{x^2}{2} - \frac{x^3}{6}$. On $0 \leq t \leq 1$ we have $x \leq b$, so $x^3 \leq b\,x^2$ and $\dot{x} \geq 1 + c\,x^2$ with $c = \frac{1}{2} - \frac{b}{6} \approx 0.2986$. Comparing with $\dot{y} = 1 + cy^2$, $y = \tan(\sqrt{c}\,t)/\sqrt{c}$:

$$x(1) \geq \frac{\tan\sqrt{c}}{\sqrt{c}} \approx 1.1131 =: a.$$

Therefore

$$\boxed{1.113 < x(1) < 1.209}.$$

(Even simpler but cruder: $1 < x(1) < \mathrm{e} - 1 \approx 1.718$, from $\dot{x} \geq 1$ and $\dot{x} \leq 1 + x$.)

(c) Numerically (high-accuracy reference), $x(1) = 1.153639\ldots \approx \mathbf{1.154}$, inside the bounds above. Euler results:

Δt	0.1	0.01	0.001	0.0001
$\hat{x}(1)$	1.12745	1.15084	1.15336	1.15361
error	2.6×10^{-2}	2.8×10^{-3}	2.8×10^{-4}	2.8×10^{-5}

The error is $\approx 0.28\,\Delta t$. For an error below 5×10^{-4} one needs $\Delta t \approx 10^{-3}$ (about 10^3 steps), and $\Delta t = 10^{-4}$ to have the printed digits agree: $1.1536 \to 1.154$. So Δt must be of order 10^{-3}–10^{-4}.

(d) With Runge–Kutta: $\Delta t = 1$: 1.153606 (error 3.3×10^{-5}); $\Delta t = 0.5$: 1.153621 (1.8×10^{-5}); $\Delta t = 0.1$: 1.15363899 (1.5×10^{-8}); $\Delta t = 0.01$: 1.153639003 (7×10^{-13}). So RK reaches three-decimal accuracy (1.154) already with $\Delta t = O(1)$ (a single step of size 1), and eight-digit accuracy with $\Delta t = 0.1$: it is orders of magnitude more efficient than Euler here, even counting its four function evaluations per step. The rigorous bounds in (b) are of course much looser than either numerical result. ■

2.8.7 *(Error estimate for Euler) Let $x_1 = x_0 + f(x_0)\Delta t$ and $x(t_1) = x(t_0 + \Delta t)$. (a) Expand $x(t_1)$ through $O(\Delta t^2)$. (b) Show that the local error is $\sim C\Delta t^2$ and find C.*

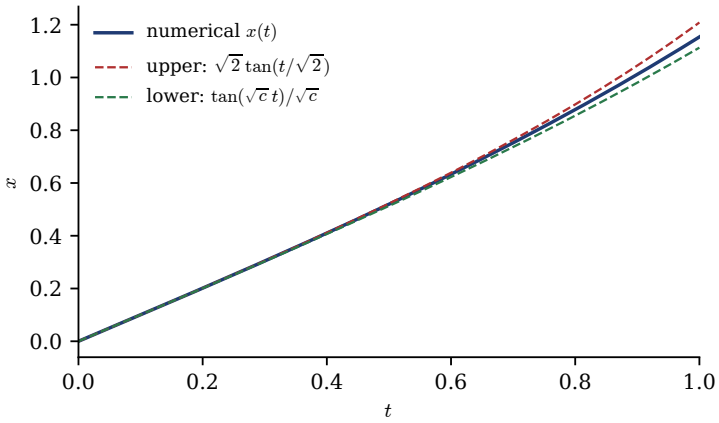


Figure 24: 2.8.6: numerical solution of $\dot{x} = x + \mathrm{e}^{-x}$ with the analytic upper and lower bounds.

Solution. **(a)** Taylor: $x(t_0 + \Delta t) = x_0 + \dot{x}(t_0)\Delta t + \frac{1}{2}\ddot{x}(t_0)\Delta t^2 + O(\Delta t^3)$. Now $\dot{x} = f(x_0)$ and $\ddot{x} = \frac{\mathrm{d}}{\mathrm{d}t} f(x) = f'(x)\dot{x} = f'(x_0)f(x_0)$, so

$$x(t_1) = x_0 + f(x_0)\Delta t + \tfrac{1}{2} f'(x_0)f(x_0)\Delta t^2 + O(\Delta t^3).$$

(b) Subtracting $x_1 = x_0 + f(x_0)\Delta t$:

$$x(t_1) - x_1 = \tfrac{1}{2} f(x_0)f'(x_0)\,\Delta t^2 + O(\Delta t^3),$$

so $|x(t_1) - x_1| \sim C\Delta t^2$ with $C = \frac{1}{2}|f(x_0)f'(x_0)|$ (generically nonzero). Over a fixed interval $T = n\Delta t$ we take $n = T/\Delta t = O(\Delta t^{-1})$ steps; each contributes an $O(\Delta t^2)$ error (and the errors propagate with bounded amplification), so the global error is $O(\Delta t^{-1}) \cdot O(\Delta t^2) = O(\Delta t)$, as claimed in the text and as observed in 2.8.3. ■

2.8.8 *(Error estimate for improved Euler) Show that the local error is $O(\Delta t^3)$.*

Solution. Write $h = \Delta t$, with $f, f', f'', \ldots$ evaluated at x_0. The step is $x_1 = x_0 + \frac{h}{2}\big[f(x_0) + f(\tilde{x})\big]$ with $\tilde{x} = x_0 + hf$. Expand

$$f(\tilde{x}) = f + f'(hf) + \tfrac{1}{2} f''(hf)^2 + O(h^3) = f + hf'f + \tfrac{1}{2} h^2 f'' f^2 + O(h^3).$$

Then

$$x_1 = x_0 + hf + \tfrac{1}{2} h^2 f'f + \tfrac{1}{4} h^3 f'' f^2 + O(h^4).$$

The exact solution: $\dot{x} = f$, $\ddot{x} = f'f$, $\dddot{x} = (f'f)'\dot{x} = (f''f + f'^2)f = f''f^2 + f'^2 f$, so

$$x(t_0 + h) = x_0 + hf + \tfrac{1}{2} h^2 f'f + \tfrac{1}{6} h^3 \big(f'' f^2 + f'^2 f\big) + O(h^4).$$

The h and h^2 terms agree. Subtracting,

$$x(t_0 + h) - x_1 = h^3 \Big[\big(\tfrac{1}{6} - \tfrac{1}{4}\big) f'' f^2 + \tfrac{1}{6} f'^2 f\Big] + O(h^4) = \frac{h^3}{12}\, f\left(2f'^2 - f f''\right) + O(h^4),$$

which is $O(\Delta t^3)$ in general (global error $O(\Delta t^2)$). Check with $f = -x$: $\frac{h^3}{12}(-x)(2) = -\frac{x\,h^3}{6}$, matching $\mathrm{e}^{-h} - (1 - h + \frac{h^2}{2}) = -\frac{h^3}{6} + \ldots$ ✓ ■

2.8.9 *(Error estimate for Runge–Kutta) Show that the local error of RK4 is $O(\Delta t^5)$ (by computer algebra).*

Solution. Let $h=\Delta t$ and write f, f', f'', f''', f'''' for f and its derivatives at x_0. For the autonomous scalar equation the RK4 stages are $k_1=f(x_0)$, $k_2=f(x_0+\tfrac{h}{2}k_1)$, $k_3=f(x_0+\tfrac{h}{2}k_2)$, $k_4=f(x_0+hk_3)$ and $x_1=x_0+\tfrac{h}{6}(k_1+2k_2+2k_3+k_4)$. Expanding each k_i in a Taylor series in h (done with sympy and checked by hand through $O(h^3)$):

$$
\begin{aligned}
k_2 &= f+\tfrac12 f'f\,h+\tfrac18 f''f^2h^2+\tfrac{1}{48}f'''f^3h^3+\cdots\\
k_3 &= f+\tfrac12 f'f\,h+\big(\tfrac14 f'^2f+\tfrac18 f''f^2\big)h^2+\big(\tfrac{3}{16}f'f''f^2+\tfrac{1}{48}f'''f^3\big)h^3+\cdots\\
k_4 &= f+f'f\,h+\big(\tfrac12 f'^2f+\tfrac12 f''f^2\big)h^2+\big(\tfrac14 f'^3f+\tfrac58 f'f''f^2+\tfrac16 f'''f^3\big)h^3+\cdots
\end{aligned}
$$

Combining gives the RK4 step

$$
\begin{aligned}
x_1-x_0 = hf+\tfrac12 h^2f'f+\tfrac16 h^3\big(f'^2f+f''f^2\big)+\tfrac{1}{24}h^4\big(f'^3f+4f'f''f^2+f'''f^3\big)\\
+\tfrac{h^5}{576}f^2\big(60f'^2f''+34f'f'''f+18f''^2f+5f''''f^2\big)+O(h^6).
\end{aligned}
$$

The exact solution, using $x^{(n+1)}=f\,\frac{\mathrm d}{\mathrm dx}x^{(n)}$, has derivatives

$$
\begin{aligned}
x'=f,\quad x''=f'f,\quad x'''=f'^2f+f''f^2,\quad x''''=f'^3f+4f'f''f^2+f'''f^3,\\
x^{(5)}=f'^4f+11f'^2f''f^2+7f'f'''f^3+4f''^2f^3+f''''f^4.
\end{aligned}
$$

Hence $x(t_0+h)=x_0+\sum_{n\ge1}x^{(n)}h^n/n!$ agrees with the RK4 step term by term through h^4: the coefficients of h,h^2,h^3,h^4 are identical (they are the "wonderful cancellations"). The first mismatch is at h^5:

$$
x(t_0+h)-x_1=\frac{h^5}{2880}\,f\Big(24f'^4-36f'^2f''f-2f'f'''f^2+6f''^2f^2-f''''f^3\Big)+O(h^6),
$$

which is generically nonzero. So the local error is $O(\Delta t^5)$ (global error $O(\Delta t^4)$). As a check, for $f=-x$ (so $f'=-1$, higher derivatives 0) the exact local error is $x(t_0+h)-x_1=x_0\big(\mathrm e^{-h}-\sum_{n\le4}(-h)^n/n!\big)=-\frac{x_0h^5}{120}+\dots$, and the formula gives $\frac{h^5}{2880}(-x_0)(24)=-\frac{x_0h^5}{120}$ ✓ ■